

Discrete Mathematics Course Map

- Binary relations
- Set theory
- Combinatorics
- Induction
- Graph theory

1 Set Theory and Functions

1.1 Elementary Set Theory

Definition 1.1

A **set** X is a well-defined collection of objects, each of which is called an element of the set:

$$X = \{x_1, x_2, x_3, \dots\} \quad (1.1)$$

Given a set X and a certain object x , one (and only one) of the following statements is true:

- the object x belongs to the set X : $x \in X$, or
- the object x does not belong to X : $x \notin X$.

The order of the elements in a set is irrelevant, as well as the number of occurrences of an element in the list.

Definition 1.2

Two sets are equal if and only if they have the same elements.

Definition 1.3

The **empty set** $\emptyset$ is the set with no elements: $\emptyset = \{\}$. The **universal set** S is the set containing all objects under consideration.

1.1.1 How to Describe a Set

- By using a roster (when it is possible to list all the elements of the set):

$$X = \{1, 2, 3, 4, 5, 6\} \quad (1.2)$$

- By using a defining predicate:

$$Y = \{y : P(y)\} \quad (1.3)$$

where $P(y)$ is a predicate containing the free variable y . Then Y is the set of all objects y such that $P(y)$ is true.

- By using set operations to build a new set from already existing sets:

$$Z = \{1, 2\} \cup \{x \in \mathbb{R} : x \in [4, 5]\} \quad (1.4)$$

- By using a recursive description of the set C in terms of another set D and some operations on the elements of D :

$$C = \{n^3 : n \in \mathbb{N}\} = \{m \in \mathbb{N} : \exists k \in \mathbb{N} \text{ such that } m = k^3\} \quad (1.5)$$

1.2 Subsets

Definition 1.4

The set A is a **subset** of the set B ($A \subseteq B$) if and only if every element of A is also an element of B . The set A is a **proper subset** of B ($A \subset B$) if A is a subset of B , and B contains at least an element not in A .

Properties:

- Every set A satisfies $A \subseteq A \subseteq S$.
- The empty set $\emptyset$ is a subset of every set A : $\emptyset \subseteq A$.

Definition 1.5

The **power set** of the set A , denoted as $\mathcal{P}(A)$, is the set of all subsets of A :

$$\mathcal{P}(A) = \{B : B \subseteq A\} \quad (1.6)$$

1.3 Set Operations

Given two sets A and B we can define the following operations:

- **Union:** $A \cup B = \{x : (x \in B) \vee (x \in A)\}$
- **Intersection:** $A \cap B = \{x : (x \in B) \wedge (x \in A)\}$
- **Complement:** $\bar{A} = \{x : x \notin A\}$, and it satisfies that $\overline{(\bar{A})} = A$
- **Difference:** $A \setminus B = \{x : (x \in A) \wedge (x \notin B)\}$
- **Symmetric difference:** $A \triangle B = \{x : (x \in A \cup B) \wedge (x \notin A \cap B)\}$

Some properties:

- **Distributive laws**
 - $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
 - $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
- **De Morgan's laws**
 - $\overline{A \cup B} = \bar{A} \cap \bar{B}$
 - $\overline{A \cap B} = \bar{A} \cup \bar{B}$
- $A \triangle B = (A \setminus B) \cup (B \setminus A)$

1.4 Cartesian Product

Definition 1.6

Given two sets X and Y , the **Cartesian product** $X \times Y$ is the set of all ordered pairs of the form:

$$X \times Y = \{(x, y) : (x \in X) \wedge (y \in Y)\} \quad (1.7)$$

Remark. $\{a, b\}$ is not the same as (a, b) . In particular, $\{1, 2\}$ is a set and therefore, $\{1, 2\} = \{2, 1\}$. However, $(1, 2)$ is an ordered pair and therefore, $(1, 2) \neq (2, 1)$.

Definition 1.7

Two sets A and B are **disjoint** if $A \cap B = \emptyset$.

1.5 Natural Numbers

Definition 1.8

The set of natural numbers $\mathbb{N}$ is defined by the following conditions:

- (i) $1 \in \mathbb{N}$.
- (ii) If $n \in \mathbb{N}$, then the successor of n (i.e., the number $n + 1$) belongs to $\mathbb{N}$.
- (iii) Every $n \in \mathbb{N}$ except 1 is the successor of some number in $\mathbb{N}$.
- (iv) Every non-empty subset of $\mathbb{N}$ has a minimum element (well-ordering property).

Notes:

- Note that $0 \notin \mathbb{N}$.
- The non-negative integers are defined as $\mathbb{Z}^+ = \{0\} \cup \mathbb{N}$.
- We can informally “define” the following sets of numbers:
 - Integer numbers: $\mathbb{Z} = \{0, \pm 1, \pm 2, \dots\}$
 - Rational numbers: $\mathbb{Q} = \left\{ \frac{p}{q} : p, q \in \mathbb{Z}, q \neq 0 \right\}$

1.6 Functions

Definition 1.9 (Spivak)

A **function** $f \subset X \times Y$ from a set X onto a set Y is a subset of the Cartesian product $X \times Y$ such that for every $x \in X$, f contains exactly one pair of the form (x, y) . The set X is called the **domain** of f and it is denoted as $\text{Dom}(f)$. The set Y is called the **codomain** of f . The **image** of f is the set

$$\text{Im}(f) = \{y : \exists x \in X \text{ such that } (x, y) \in f\} \quad (1.8)$$

Given two sets X and Y , a function is an object that assigns to each element $x \in X$ a unique element $y \in Y$, which is denoted as $y = f(x)$. Usually, functions are denoted as $f : X \rightarrow Y$.

1.7 Function Types

Definition 1.10

Given a function $f : X \rightarrow Y$, we say that

- f is **injective** if $x_1 \neq x_2$ implies $f(x_1) \neq f(x_2)$.
- f is **surjective** if for every $y \in Y$, there exists at least an element $x \in X$ such that $y = f(x)$.
- f is **bijective** if it is injective and surjective.

If $f : X \rightarrow Y$ is bijective, we can define its **inverse function** $f^{-1} : Y \rightarrow X$ by the well-defined rule:

$$f^{-1}(y) = x \iff y = f(x) \quad (1.9)$$

Given two functions $f : X \rightarrow Y$, $g : Y \rightarrow Z$, we can define a new function $g \circ f : X \rightarrow Z$ by the following rule:

$$(g \circ f)(x) = g(f(x)) \quad (1.10)$$

The function $g \circ f$ is the **composition** of f and g .

2 Elementary Combinatorics I

Definition 2.1

Let S be a set. If there are exactly $n \in \mathbb{N}$ distinct elements in S , we say that S is a **finite set**, and that n is the **cardinality** of S . The cardinality of S is denoted by $|S|$.

Definition 2.2

Two sets A and B have the same cardinality if and only if there exists a **bijective function** $f : A \rightarrow B$.

Definition 2.3

A set that is either finite or has the same cardinality as the set $\mathbb{N}$ is called **countable**.

The **goal of combinatorics** is to compute the cardinality of certain finite sets.

2.1 Basic Counting Principles

- (i) **The sum rule:** if $A \cap B = \emptyset$, then $|A \cup B| = |A| + |B|$.
- (ii) **The product rule:** $|A \times B| = |A| \cdot |B|$.
- (iii) **The inclusion-exclusion principle:** $|A \cup B| = |A| + |B| - |A \cap B|$.
- (iv) **The pigeonhole principle.** (See problem set.)

2.2 The Sum Principle

Proposition 2.4 (The sum principle v1)

If A and B are two finite and disjoint sets $A \cap B = \emptyset$, then

$$|A \cup B| = |A| + |B| \quad (2.1)$$

Proposition 2.5 (The sum principle v2)

If $A_1, A_2, \dots, A_m$ are a sequence of finite and pairwise disjoint sets $A_i \cap A_j = \emptyset$ for all $i \neq j$, then

$$|A_1 \cup A_2 \cup \dots \cup A_m| = |A_1| + |A_2| + \dots + |A_m| = \sum_{j=1}^m |A_j| \quad (2.2)$$

Proposition 2.6 (The sum principle v3)

If a first task can be done in n_1 ways, and a second task in n_2 ways, and if these tasks cannot be done at the same time, then there are $n_1 + n_2$ ways to do either task.

2.3 The Product Principle

Proposition 2.7 (The product principle v1)

If A and B are two finite sets, then

$$|A \times B| = |A| \cdot |B| \quad (2.3)$$

Proposition 2.8 (The product principle v2)

If $A_1, A_2, \dots, A_m$ are finite sets, then

$$|A_1 \times A_2 \times \dots \times A_m| = |A_1| \cdot |A_2| \dots |A_m| = \prod_{k=1}^m |A_k| \quad (2.4)$$

Proposition 2.9 (The product principle v3)

Suppose that a procedure can be broken down into two tasks. If there are n_1 ways to perform the first task, and n_2 ways to perform the second task after the first task has been done, then there are $n_1 \cdot n_2$ ways to do the procedure.

2.4 Permutations

Definition 2.10

For each positive integer $n \in \mathbb{N}$, we define the **factorial** of n as

$$n! = n \cdot (n - 1) \cdot (n - 2) \dots 2 \cdot 1 \quad (2.5)$$

Proposition 2.11 (Permutations of n distinct objects)

Given n distinct objects, there are $n!$ distinct ordered arrangements (= permutations) of these objects.

Proposition 2.12 (Permutations with repetition)

Given n objects that can be classified into k groups of identical objects, and such that the first group contains n_1 identical elements, the second group contains n_2 identical elements, etc., then the number of distinct ordered arrangements of these objects is

$$\binom{n}{n_1, n_2, \dots, n_k} \equiv \frac{n!}{n_1! n_2! \dots n_k!} \quad (2.6)$$

with $\sum_{i=1}^k n_i = n$.

2.5 Ordered Subsets

Proposition 2.13

Given a set of n distinct elements, we can form

$$n(n-1)(n-2)\dots(n-r+1) = \frac{n!}{(n-r)!} \quad (2.7)$$

ordered subsets containing r elements.

Remark. If $r = n$, the first formula implies that there are $n!$ permutations of a set of n distinct elements. The second formula only makes sense if we define $0! = 1$.

Proposition 2.14

The number of ordered subsets of r elements taken from a set of n distinct objects with repetition allowed is n^r .

2.6 Subsets

Proposition 2.15

The number of distinct subsets with r elements that can be extracted from a set of n distinct elements is given by:

$$\binom{n}{r} = \frac{n!}{r!(n-r)!} \quad (2.8)$$

The symbol $\binom{n}{r}$ is read “ n choose r ”.

Definition 2.16 (Binomial coefficients)

For all non-negative integers $n, r \in \mathbb{Z}^+$ such that $0 \leq r \leq n$, we define the **binomial coefficient** $\binom{n}{r}$ as follows:

$$\binom{n}{r} = \frac{n!}{r!(n-r)!} \quad (2.9)$$

where we define $0! = 1$.

Remark. $\binom{n}{k} = 0$ whenever $k < 0$, or $k > n$.

2.7 Binomial Coefficients: Pascal's Triangle

$$\begin{array}{ccccccc}
 & & & & 1 & & \\
 & & & 1 & & 1 & \\
 & & 1 & & 2 & & 1 \\
 & 1 & & 3 & & 3 & & 1 \\
 1 & & 4 & & 6 & & 4 & & 1
 \end{array}$$

$$\binom{n}{r} \equiv \binom{\text{row}}{\text{column}} \quad (2.10)$$

Theorem 2.17 (Symmetry)

$$\binom{n}{r} = \binom{n}{n-r} = \frac{n!}{r!(n-r)!}, \quad n \geq 0, 0 \leq r \leq n \quad (2.11)$$

Theorem 2.18 (Pascal's identity)

$$\binom{n+1}{r} = \binom{n}{r} + \binom{n}{r-1}, \quad n \geq 0, 0 < r \leq n \quad (2.12)$$

2.8 Newton's Binomial Theorem

Theorem 2.19 (Newton's binomial theorem)

$$(x+y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k}, \quad n \geq 0 \quad (2.13)$$

Corollary 2.19.1

$$(1+x)^n = \sum_{k=0}^n \binom{n}{k} x^k, \quad n \geq 0 \quad (2.14)$$

Corollary 2.19.2

For every $n \geq 0$,

$$\sum_{k=0}^n \binom{n}{k} = 2^n, \quad \sum_{k=0}^n (-1)^k \binom{n}{k} = 0 \quad (2.15)$$

2.9 Cardinality of the Power Set

Corollary 2.19.3

Given a finite set A , then

$$|\mathcal{P}(A)| = 2^{|A|} \quad (2.16)$$

2.10 The Inclusion-Exclusion Principle

Proposition 2.20 (The inclusion-exclusion principle v1)

$$|A \cup B| = |A| + |B| - |A \cap B| \quad (2.17)$$

Proposition 2.21 (The inclusion-exclusion principle v2)

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C| \quad (2.18)$$

Proposition 2.22 (The inclusion-exclusion principle v3)

$$|A_1 \cup A_2 \cup \dots \cup A_n| = \sum_{1 \leq i \leq n} |A_i| - \sum_{1 \leq i < j \leq n} |A_i \cap A_j| \quad (2.19)$$

$$+ \sum_{1 \leq i < j < k \leq n} |A_i \cap A_j \cap A_k| - \dots \quad (2.20)$$

$$+ (-1)^{n+1} |A_1 \cap A_2 \cap \dots \cap A_n| \quad (2.21)$$

Proposition 2.23 (The inclusion-exclusion principle v4)

Given sets $A_i \subset S$ with $1 \leq i \leq n$, then

$$|\overline{A_1 \cup A_2 \cup \dots \cup A_n}| = |\overline{A_1} \cap \overline{A_2} \cap \dots \cap \overline{A_n}| = |S| - |A_1 \cup A_2 \cup \dots \cup A_n| \quad (2.22)$$

Remarks.

- $\overline{A_1} \cap \overline{A_2} \cap \dots \cap \overline{A_n} = \{x : x \notin A_1, x \notin A_2, \dots, x \notin A_n\}$
- $\overline{A} = S \setminus A \Rightarrow |\overline{A}| = |S| - |A|$

3 Elementary Combinatorics II

3.1 Basic Counting Principles (Review)

- (i) **The sum rule:** if $A \cap B = \emptyset$, then $|A \cup B| = |A| + |B|$.
- (ii) **The product rule:** $|A \times B| = |A| \cdot |B|$.
- (iii) **The inclusion-exclusion principle:** $|A \cup B| = |A| + |B| - |A \cap B|$.
- (iv) **The pigeonhole principle.**
- (v) **Other standard counting problems:**
 - Distributions.
 - Partitions.

3.2 Distributions

Proposition 3.1 (Distributions)

The number of distributions of a given set of identical r objects into n (distinct) groups, and such that each group contains at least one object, is given by $\binom{r-1}{n-1}$.

Proposition 3.2

The number of distributions of a given set of identical r objects into n (distinct) groups is given by $\binom{n+r-1}{r}$.

3.3 Set Partitions

Definition 3.3

Let S be a finite set of cardinality n . A **partition** of S of type $(n_1, n_2, \dots, n_k)$ with $n_i \in \mathbb{N}$ is the set $\{S_i\}_{i=1}^k$, where the subsets S_i satisfy: (1) $|S_i| = n_i$ for all $1 \leq i \leq k$, (2) are pairwise disjoint: $S_i \cap S_j = \emptyset$ for all $i \neq j$; and (3) their union is S (therefore, $\sum_{i=1}^k n_i = n$).

Proposition 3.4

Let S be a set of cardinality $m \cdot n$. Then, there exist

$$\frac{(m \cdot n)!}{(m!)^n n!} \quad (3.1)$$

distinct partitions of S into n subsets S_i of type $(m, m, \dots, m)$.

Proposition 3.5

The number of distinct partitions of a set of cardinality m of type $(m_1, m_2, \dots, m_n)$ is given by

$$\binom{m}{m_1, m_2, \dots, m_n} \prod_{k \geq 1} \frac{1}{r_k!} \quad (3.2)$$

where r_k is the number of subsets of cardinality k .

4 Graph Theory I

4.1 Undirected Graphs

(i) Undirected graphs:

- Basic notation and definitions.
- Graph representation.
- Graph isomorphism.
- Walks in a graph.

- Trees.
 - Planar graphs.
- (ii) **Algorithms in graph theory.**
- (iii) **Combinatorial problems on graphs.**

Definition 4.1

A **pseudograph** $G = (V, E, \gamma)$ consists of a nonempty vertex set V , an edge set E , and a function $\gamma : E \rightarrow \{\{u, v\} : u, v \in V\}$

- The function γ encodes the graph connectivities.
- If $e \in E$ satisfies $\gamma(e) = \{u, v\}$ with $u \neq v$, we say that u and v are **adjacent**, and that e is **incident** with u and v .
- If there two distinct edges $e_1, e_2 \in E$ such that $\gamma(e_1) = \gamma(e_2) = \{a, b\}$, then we say that e_1 and e_2 are **multiple edges**.
- If there exists $e \in E$ such that $\gamma(e) = \{v, v\} = \{v\}$, then e is a **loop** incident with v .
- Hereafter, if we do not say it explicitly, we will assume that $G = (V, E)$ is undirected.

Definition 4.2

A **multigraph** $G = (V, E)$ is a pseudograph in which multiple edges are allowed, but loops are not allowed. A **simple graph** $G = (V, E)$ is a pseudograph in which loops and multiple edges are not allowed.

4.2 More Definitions

Definition 4.3

The **degree** (or valence) of a vertex $v \in V$ in a graph $G = (V, E)$ is the number of edges incident with it, except that a loop contributes twice to the degree of that vertex. The degree of a vertex v is denoted by $d(v)$.

Remark. Given a vertex $v \in V$, its degree $d(v)$ is equal to

$$d(v) = |\{\{v, y\} \in E : y \neq v\}| + 2 \times \text{number of loops incident with } v \quad (4.1)$$

Definition 4.4

A vertex of degree 1 is called a **terminal** (or a **pendant vertex**). A vertex of degree 0 is called an **isolated vertex**. A graph with no edges is called **trivial**.

Definition 4.5

A **regular graph** is a graph such that all vertices have the same degree.

4.3 The Handshaking Theorem

Theorem 4.6 (The Handshaking Theorem)

In any undirected graph $G = (V, E)$, we have that

$$\sum_{v \in V} d(v) = 2|E| \quad (4.2)$$

Corollary 4.6.1

For any graph G , the sum of all vertex degrees is an even number.

Theorem 4.7

Any graph has an even number of vertices of odd degree.

Corollary 4.7.1

For any graph G with an odd number of vertices, there is an odd number of vertices of even degree.

4.4 More Definitions

Definition 4.8

A graph $G = (V, E)$ is **bipartite** if its vertex set V can be partitioned into two disjoint subsets V_1 and V_2 such that every edge in the graph connects a vertex in V_1 with a vertex in V_2 .

Simple graph families:

- The complete graph on n vertices K_n .
- The path P_n on n vertices.
- The cycle C_n on n vertices.
- The wheel graph on $n + 1$ vertices W_n .
- The complete bipartite graph on n and m vertices $K_{n,m}$.
- The n -cube graphs Q_n are defined as follows: each vertex represents a bit string of length n , and two vertices u and v are adjacent if and only if the corresponding bit strings differ in exactly one bit.

4.5 Complementary Graph and Subgraphs

Definition 4.9

The **complementary graph** $\overline{G} = (V, \overline{E})$ of a simple graph $G = (V, E)$ has the same vertex set as G , and two vertices are adjacent in $\overline{G}$ if and only if they are not adjacent in G .

Definition 4.10

The graph $H = (W, F)$ is a **subgraph** of $G = (V, E)$ if $W \subseteq V$ and $F \subseteq E$.

Definition 4.11

Given a graph $G = (V, E)$, a **spanning subgraph** of G is any subgraph $H = (V, F)$ of G (hence, $F \subseteq E$).

Definition 4.12

Let $G = (V, E)$ be a graph, and $v_1, v_2, \dots, v_{|V|}$ be a fixed ordering of its vertex set V . The **adjacency matrix** of G associated to that particular vertex ordering is the matrix of dimensions $|V| \times |V|$ such that its entry A_{ij} counts the number of edges joining the vertices v_i and v_j .

4.6 Isomorphism of Graphs

Remark. Do not confuse a graph with its graphical representation!

Definition 4.13

The simple graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ are **isomorphic** if and only if there exist a bijective function $f : V_1 \rightarrow V_2$ with the following property: a and b are adjacent in G_1 if and only if $f(a)$ and $f(b)$ are adjacent in G_2 . The function f is called an **isomorphism**.

Remark. Given two simple graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$, then

- (i) If $|V_1| \neq |V_2|$, then G_1 and G_2 are not isomorphic.
- (ii) If $|E_1| \neq |E_2|$, then G_1 and G_2 are not isomorphic.
- (iii) If S_i is the degree sequence of the graph G_i , and $S_1 \neq S_2$, then G_1 and G_2 are not isomorphic.
- (iv) Other methods.

Remark. G_1 and G_2 are isomorphic if there exists an invertible linear map (basically a permutation of the “basis vectors”) $\pi : V_1 \rightarrow V_2$ such that $A_2 = P^{-1} \cdot A_1 \cdot P$. There are $|V_1|! = |V_2|!$ maps of this type!

4.7 Walks in a Graph

Definition 4.14

A **walk** in a graph $G = (V, E)$ is an alternating sequence of vertices and edges of the form $v_0, \{v_0, v_1\}, v_1, \{v_1, v_2\}, v_2, \dots, v_{\ell-1}, \{v_{\ell-1}, v_\ell\}, v_\ell$. The length of the walk is equal to the number of edges in the walk. There is an implicit direction in every walk: v_0 is the initial vertex, and v_ℓ is the final vertex.

Definition 4.15

A **trail** is a walk in which no edge occurs more than once. A closed trail is called a **circuit**. A **path** is a trail in which all of its vertices are different. A **cycle** is a closed path of positive length.

Remark. Circuit can be also used as a synonym of closed walk, or closed path (cycle).

4.8 Number of Walks Between Two Vertices

Theorem 4.16

Let G be a graph with adjacency matrix A with respect to the ordering $\{v_1, v_2, \dots, v_{|V|}\}$ of its vertex set. The number of distinct oriented walks of length $n \geq 1$ that start at v_i and end at v_j is given by the entry (i, j) of the matrix A^n .

Corollary 4.16.1

Let G be a simple graph with adjacency matrix A , then

- $A_{ii}^2 = d(i)$ for every $1 \leq i \leq |V|$.
- $\text{tr } A^2 = 2|E|$.
- $\text{tr } A^3 = 6 \times \text{number of unoriented triangles in } G$.

4.9 Connected Graphs

Definition 4.17

An undirected graph is **connected** if there is a path between every pair of distinct vertices of G . A **disconnected** graph is formed by the disjoint union of several connected subgraphs called the **connected components** of the graph.

Remark. If two vertices of a graph can be connected by a walk, then there is at least one path connecting them. These paths correspond to the walks of minimum length connecting these two vertices.

Definition 4.18

An **articulation point** or **cut vertex** of a graph G is a vertex whose removal (together with those edges incident with it) produces a graph with more connected components than in G . A **cut edge** or **bridge** of a graph G is an edge whose removal produces a graph with more connected components than in G .

5 Graph Theory II

5.1 Trees

- (i) **Undirected graphs:**

- Basic notation and definitions.
 - Graph representation.
 - Graph isomorphism.
 - Walks in a graph.
 - **Trees.**
 - Planar graphs.
- (ii) **Algorithms in graph theory.**
- (iii) **Combinatorial problems on graphs.**

Definition 5.1

A **tree** is a simple connected graph with no cycles. A **forest** is a simple graph with no cycles. Each connected component of a forest is a tree.

Remark. Trees may be rooted trees. A rooted tree is a tree with one distinguished vertex (the root). Hereafter, we will assume that all trees are rootless, unless specified.

Theorem 5.2

- (i) The simple graph G is a tree if and only if it is connected and, if we remove any edge, we obtain a disconnected graph.
- (ii) The simple graph G is a tree if and only if it does not contain any cycles and, if we add any edge, we create a cycle.

Theorem 5.3

A graph $G = (V, E)$ is a tree if and only if there exists a unique path between any pair of vertices.

Theorem 5.4

Any tree with at least two vertices contains at least two vertices of degree one.

5.2 Properties of Trees

Definition 5.5

How to grow a tree?

- (i) Start from the trivial tree $T = (\{r\}, \emptyset)$, where r is the root vertex.
- (ii) Given $T = (V, E)$, add a new vertex u and a new edge $\{u, v\}$ where $v \in V$.

Theorem 5.6

Any graph obtained by using the preceding procedure is a tree, and any tree can be obtained in this way.

Theorem 5.7

Any tree with n vertices has $n - 1$ edges.

Theorem 5.8

If G is a graph with n vertices, then the following statements are equivalent:

- (i) G is a tree.
- (ii) G is connected and has $n - 1$ edges.
- (iii) G has $n - 1$ edges and does not contain any cycle.

5.3 Planar Graphs

Definition 5.9

A **planar graph** is a graph that can be embedded in the plane: i.e., it can be drawn on the plane in such a way that their edges do not cross each other. A **plane graph** is a graphical representation of a planar graph such that their edges do not cross each other.

Definition 5.10

A **subdivision** of an edge results from inserting a new vertex into that edge. The subdivision of a graph G is obtained by subdividing one or more edges in G .

Theorem 5.11 (Kuratowski, 1930)

A graph is planar if and only if it does not contain a subgraph that is a subdivision of K_5 or $K_{3,3}$.

5.4 Planar and Dual Graphs

Theorem 5.12 (Euler's formula, 1752)

A plane and connected graph $G = (V, E)$ divides the plane into R regions (or faces), such that

$$|V| - |E| + R = 2 \quad (5.1)$$

A plane graph (not necessarily connected) divides the plane into R regions, such that

$$|V| - |E| + R = 1 + \text{number of connected components of } G \quad (5.2)$$

Definition 5.13

Given a plane connected graph $G = (V, E)$, we can define its **dual graph** $G^* = (V^*, E^*)$ in the following way: To each region f of G we associate a dual vertex $f^* \in V^*$, and to

each edge $e \in E$, there corresponds a unique dual edge $e^* \in E^*$. If the original edge e is the intersection of two faces f, h (possibly, $f = h$), then the corresponding dual edge e^* is incident with the dual vertices $f^*, g^* \in V^*$.

- G^* can be drawn in such a way that any dual edge e^* only crosses e .
- Notice that $(G^*)^* = G$.

5.5 Some Corollaries About Graph Planarity

Definition 5.14

Given a plane graph, the **degree of a region** r is the degree of the dual vertex $r \in V^*$ associated with it in the dual graph G^* . We denote the degree of the region r as d_r (or $d(r)$).

Theorem 5.15 (Handshake Theorem for the dual graph)

Given a plane connected graph G , then

$$2|E| = \sum_{r \in R} d_r \quad (5.3)$$

where R is the set of regions defined on the plane by G .

Corollary 5.15.1

If G is a simple, connected, and planar graph with $|V| \geq 3$, then $|E| \leq 3|V| - 6$.

Corollary 5.15.2

If G is a simple, connected, and planar graph with $|V| \geq 3$ and without cycles of length 3, then $|E| \leq 2|V| - 4$.

6 Graph Theory III

6.1 Algorithms in Graph Theory

- (i) **Undirected graphs.**
- (ii) **Algorithms in graph theory:**
 - **Minimum-weight spanning tree:** Prim's and Kruskal's algorithms.
 - **Shortest path:** Dijkstra's algorithm.
 - **Graph colorings.**
 - **Eulerian and Hamiltonian graphs.** Fleury's algorithm.
- (iii) **Combinatorial problems on graphs.**

6.2 Minimum-Weight Spanning Tree

Definition 6.1

A **spanning tree** of a connected graph G is a subgraph of G that is a tree and contains all vertices of G .

Definition 6.2

A **weighted graph** $G = (V, E, \omega)$ is a graph such that every edge $e \in E$ is associated to a weight $\omega(e) \in \mathbb{R}$.

Definition 6.3

A **minimum-weight spanning tree** of a connected weighted graph $G = (V, E, \omega)$ is a spanning tree $T = (V, A)$ of G such that $\omega(A) = \sum_{e \in A} \omega(e)$ takes the minimum possible value.

Problem 1: Find a minimum-weight spanning tree of a connected weighted graph $G = (V, E, \omega)$.

Remark. The number of trees on n vertices grows very rapidly with n .

Definition 6.4

A **greedy algorithm** to solve a given problem is an algorithm such that at every step, it always takes, among all the choices allowed by the problem, the optimum one.

6.3 Prim's Algorithm, 1957

Algorithm 6.1: Prim's algorithm

```
1  $T_1 = (V_1, E_1)$  where  $E_1 = \{e_1\}$ ,  $e_1 = \{x_0, x_1\}$  is one edge with minimum weight  $\omega_{\min}$ ,  
   and  $V_1 = \{x_0, x_1\}$ .  
2 for  $i = 1$  to  $n - 2$   
3    $e_{i+1} = \{x_i, x_{i+1}\}$  edge of minimum weight that is incident with a vertex  $x_j$  of  $T_i =$   
    $(V_i, E_i)$ , and such that it does not form a cycle when added to  $T_i$   
4    $T_{i+1} = (V_i \cup \{x_{i+1}\}, E_i \cup \{e_{i+1}\}) = (V_{i+1}, E_{i+1})$ 
```

Remarks.

- The edge e_i ($i = 1, \dots, n - 1$) might not be unique.
- The minimum-weight spanning tree might not be unique.
- At each step, T_i is a tree ($1 \leq i \leq n - 1$).

Theorem 6.5

Given a connected weighted graph $G = (V, E, \omega)$, Prim's algorithm produces a minimum-weight spanning tree of G .

6.4 Kruskal's Algorithm, 1957

Algorithm 6.2: Kruskal's algorithm

```

1  $T_0 = (V, E_0)$  with  $E_0 = \emptyset$ 
2 for  $i = 1$  to  $n - 1$ 
3    $e_i =$  edge of minimum weight such that it does not form a cycle when added to  $T_{i-1} =$ 
    $(V, E_{i-1})$ 
4    $T_i = (V, E_{i-1} \cup \{e_i\}) = (V, E_i)$ 

```

Remarks.

- The edge e_i ($i = 1, \dots, n - 1$) might not be unique.
- At each step, T_i is a forest ($1 \leq i \leq n - 1$).

Theorem 6.6

Given a connected weighted graph $G = (V, E, \omega)$, Kruskal's algorithm produces a minimum-weight spanning tree of G .

6.5 Shortest Path Between Two Vertices: Dijkstra's Algorithm, 1959

Problem 2: Find the shortest path that joins an initial vertex s to a final vertex t belonging to a simple, connected, and weighted graph $G = (V, E, \omega)$ such that all weights are positive ($\omega_e > 0$ for every edge $e \in E$).

Theorem 6.7

Dijkstra's algorithm finds the length of the shortest path between two vertices of a simple, connected, and weighted graph $G = (V, E, \omega)$ with all its weights being positive.

The basic idea: In each iteration, we assign to each vertex j two labels, that might be either temporary (δ_j, P_j) or permanent (δ_j, P_j) .

- The label δ_j is an estimate of the length of the path going from the initial vertex s to the vertex j .
- The label P_j is an estimate of the predecessor of the vertex j along the above path.

We will denote the weight of the edge $\{i, j\} \in E$ as $\omega(\{i, j\}) = \omega_{ij} > 0$.

6.6 Dijkstra's Algorithm

- Initial Step:** We mark the origin s with the permanent label $(0, s)$. All the other vertices $j \in V$ ($j \neq s$) are marked with temporary labels:

- If $\{j, s\} \in E$, we assign the label $(\omega_{s,j}, s)$ to j .
 - If $\{j, s\} \notin E$, we assign to j the label $(\infty, -)$.
- (ii) Let $v \in V$ be the last vertex that has become permanent. For each temporary vertex j , we compare the temporary label δ_j to the new value $\delta_v + \omega_{v,j}$:
- If $\delta_v + \omega_{v,j} < \delta_j$, the old label (δ_j, P_j) is replaced by $(\delta_v + \omega_{v,j}, v)$.
 - If $\delta_v + \omega_{v,j} > \delta_j$, the label (δ_j, P_j) remains the same.
- (iii) Among all temporary vertices j , we choose one j_0 with the minimum label
- $$\delta_{j_0} = \min(\delta_j : j \text{ is temporary}) = \delta_{\min} \quad (6.1)$$
- If $\delta_{\min} = \infty$, the algorithm ends: there is no path between s and t .
 - If $\delta_{\min} < \infty$, we mark such vertex with the permanent label $(\delta_{\min}, P_{j_0})$.
- (iv) If t is the vertex whose label has become permanent, the algorithm ends. The length of the shortest path between s and t is δ_t , and such a path is obtained by following the permanent labels in reverse order $t \rightarrow P_t \rightarrow \dots \rightarrow s$. Otherwise, go back to Step (2).

Remarks.

- If at a given step there are several options, we can choose any of them.
- The shortest path between two vertices may not be unique; but the shortest length does not depend on such a choice.
- The final output of Dijkstra's algorithm (when all vertices have been marked permanent) is a rooted spanning tree T , such that the root is the initial vertex s , and the distance between s and any other vertex j of the graph is the sum of all the weights of the unique path between s and j on T .

6.7 Directed Graphs or Digraphs

Definition 6.8

A **directed graph** $G = (V, E)$ consists in a nonempty set of vertices V and an edge set E , such that each edge $e \in E$ is an ordered pair of vertices $e = (x, y)$ with $x, y \in V$.

Definition 6.9

Let G be a directed graph $G = (V, E)$, and let $v \in V$ be a vertex of G . The **indegree** $d_i(v)$ of v is the number of edges whose second entry is v . The **outdegree** $d_o(v)$ of v is the number of edges whose first entry is v .

Proposition 6.10

In any directed graph $G = (V, E)$:

$$\sum_{v \in V} d_i(v) = \sum_{v \in V} d_o(v) = |E| \quad (6.2)$$

Definition 6.11

Let $G = (V, E)$ be a directed graph, and we consider the ordering $v_1, v_2, \dots, v_{|V|}$ of its vertex set V . The **adjacency matrix** of G associated to that ordering is the $|V| \times |V|$ matrix whose entries A_{ij} count the number of edges (v_i, v_j) that start at v_i and end at v_j .

Definition 6.12

A **walk** of length ℓ in a directed graph $G = (V, E)$ is a sequence of ℓ edges of the form $(v_0, v_1), (v_1, v_2), \dots, (v_{\ell-1}, v_\ell)$.

The definitions of trail, path, closed walk, circuit, and cycle are the natural generalization of those given for undirected graphs in Chapter 4.

We can also define weighted directed graphs $G = (V, E, \omega)$ in an analogous way.

7 Graph Theory IV

7.1 Graph Colorings

- (i) **Undirected graphs.**
- (ii) **Algorithms in graph theory:**
 - Minimum-weight spanning tree: Prim's and Kruskal's algorithms.
 - Shortest path: Dijkstra's algorithm.
 - **Graph colorings.**
 - Eulerian and Hamiltonian graphs. Fleury's algorithm.
- (iii) **Combinatorial problems on graphs.**

7.2 Proper Colorings of a Graph

Definition 7.1

A **proper coloring** (with q colors) of a simple graph $G = (V, E)$ is a function $c : V \rightarrow \{1, 2, \dots, q\}$ such that $c(u) \neq c(w)$ whenever u and w are adjacent.

- Given a graph $G = (V, E)$, the total number of vertex colorings (including both proper and improper colorings) with q colors is $q^{|V|}$.
- Hereafter, we will consider proper colorings.
- **Two difficult questions:**
 - (i) How many distinct colorings with q colors $P_G(q)$ can be obtained from a graph G ?
 - (ii) Which is the minimum number of colors q needed to color a given graph G ?

Definition 7.2

The **chromatic number** $\chi(G)$ of a graph G is the minimum positive integer q such that there is at least one coloring of G with q colors; i.e., $P_G(q) > 0$ for every $q \geq \chi(G) \in \mathbb{N}$.

Remark. Deciding whether an arbitrary graph G can be colored or not with $k \geq 3$ colors is a hard problem.

7.3 Greedy Algorithm for Coloring a Graph

Algorithm 7.3: Greedy algorithm

```
1 We order the vertices of  $V$ :  $(v_1, v_2, \dots, v_n)$ 
2  $c(v_1) = 1$ 
3 for  $i = 2$  to  $n$ 
4    $S_i = \{q : c(v_k) = q, \text{ for every } v_k \text{ that is adjacent to } v_i \text{ with } k < i\}$ 
5    $c(v_i) = \min(\overline{S_i} \cap \mathbb{N}) = \text{the smallest color not in } S_i$ 
```

Remarks.

- This algorithm does not compute $\chi(G)$, but an upper bound of $\chi(G)$ which depends (strongly) on the chosen vertex ordering.
- To compute the value of $\chi(G)$, we should consider the $n!$ possible orderings of the n vertices of G (= exponential time!).

7.4 Some Theorems

Theorem 7.3

If G is a graph with maximum degree k , then $\chi(G) \leq k + 1$.

Theorem 7.4 (Brooks, 1941)

If G is a connected non-complete graph with maximum degree $k \geq 3$, then $\chi(G) \leq k$.

Proposition 7.5

A graph G is bipartite if and only if $\chi(G) = 2$.

Theorem 7.6

A graph is bipartite if and only if it does not contain any cycle of odd length.

Corollary 7.6.1

Any tree is bipartite.

Theorem 7.7 (The four-color theorem, Appel and Haken, 1976)

For any planar graph G , $P_G(4) > 0$.

- The original proof was computer assisted.

- There is no three-color theorem: there are planar graphs G with chromatic number $\chi(G) = 4$: e.g. K_4 .

7.5 Eulerian Graphs

Problem 4 (Euler): The old city of Königsberg was crossed by a river and there were seven bridges. Was it possible to start walking at some point of the city and get back to the same place by crossing every bridge exactly once?

Problem 5: Given a graph $G = (V, E)$, is there any circuit containing every edge $e \in E$? (If it is a circuit, then each edge is visited exactly once).

Definition 7.8

An **Euler tour** is a circuit containing every edge of the graph. A graph admitting an Euler tour is an **Eulerian graph**.

An **Euler trail** is an open trail that contains all the edges of the graph.

Theorem 7.9

A connected graph is Eulerian if and only if the degree of all its vertices is even. A connected graph contains an Euler trail if and only if it contains exactly two vertices of odd degree. A connected and directed graph is Eulerian if and only if for every vertex $v \in V$, $d_i(v) = d_o(v)$.

Therefore, the **problem of the bridges of Königsberg** does not have any solution: the corresponding graph does not admit any Euler tour/trail.

7.6 Fleury's Algorithm

Let $G = (V, E)$ be a connected graph with all its vertices of even degree:

- (i) **Initial step:** We choose any vertex v_0 as the initial vertex of the Euler tour $C_0 = (v_0)$ and we define $G_0 = (V_0, E_0) = G$. The algorithm sequentially increases the tour C_0 , while it sequentially deletes elements from G .
- (ii) **How to extend the trail?:** Let $C_i = (v_0, e_1, v_1, \dots, e_i, v_i)$ be the trail corresponding to the graph $G_i = (V_i, E_i) \subseteq G_0$.
 - If there exists a unique edge incident with v_i , $e_{i+1} = \{v_i, w\} \in E_i = E \setminus \{e_1, e_2, \dots, e_i\}$:
 - $C_{i+1} = (v_0, e_1, v_1, \dots, e_i, v_i, e_{i+1}, w)$.
 - $G_{i+1} = (V_i \setminus \{v_i\}, E_i \setminus \{e_{i+1}\}) = (V_{i+1}, E_{i+1})$.
 - If there are several edges in E_i incident with v_i , we can choose any of these edges as long as the chosen one is not a bridge of G_i . If we choose $e_{i+1} = \{v_i, w\} \in E_i$:
 - $C_{i+1} = (v_0, e_1, v_1, \dots, e_i, v_i, e_{i+1}, w)$.
 - $G_{i+1} = (V_i, E_i \setminus \{e_{i+1}\}) = (V_{i+1}, E_{i+1})$.
- (iii) We repeat Step (2) $|E|$ times until $G_{|E|} = (\emptyset, \emptyset)$. Then $C_{|E|}$ is the Euler tour we were looking for.

7.7 Hamiltonian Graphs

Problem 6: Is it possible to find a cycle on a graph G such that it contains all vertices of G exactly once?

Definition 7.10

A **Hamilton cycle** of a graph G is a cycle that contains all the vertices of G . A graph admitting one Hamilton graph is a **Hamiltonian graph**.

A **Hamilton path** of a graph G is an open path that contains all vertices of G .

The problem of deciding that a given graph is Hamiltonian or not is hard.

Theorem 7.11 (Dirac, 1950)

If G is a simple graph with $n \geq 3$ vertices and each vertex has a degree $\geq n/2$, then G is a Hamiltonian graph.

Remark. Not every Hamiltonian graph satisfies the above condition: e.g. C_n with $n \geq 5$.

8 Advanced Methods in Combinatorics

8.1 Recurrence Relations

(i) **Recurrence relations:**

- Definitions.
- Solution of a linear homogeneous recurrence relation.
- Solution of a linear nonhomogeneous recurrence relation.

(ii) **Generating functions.**

Definition 8.1

A **recurrence relation** for the sequence $(a_n)_{n \in \mathbb{N}}$ is an equation that expresses a_n in terms of one or more terms $a_{n-1}, a_{n-2}, \dots, a_{n-k}$, and possibly n . In other words, it is, for any fixed $k \geq 1$, an equation of the type

$$F(n; a_n, a_{n-1}, a_{n-2}, \dots, a_{n-k}) = 0 \quad (8.1)$$

which is valid for all $n \geq k + 1$. The **initial conditions** are the first k terms in the sequence: i.e., $(a_1, \dots, a_k)$.

Definition 8.2

- A recurrence relation is of **k -th order** if a_n can be expressed in terms of k terms $a_{n-1}, a_{n-2}, \dots, a_{n-k}$.
- A recurrence relation is **linear** if it expresses a_n as a linear function of $a_{n-1}, a_{n-2}, \dots, a_{n-k}$. Otherwise, the relation is **nonlinear**.

- A recurrence relation is **homogeneous** if the zero sequence $a_n = a_{n-1} = \dots = a_{n-k} = 0$ satisfies the relation. Otherwise, it is **nonhomogeneous**.

8.2 Solution of a Linear Homogeneous Recurrence Relation

Theorem 8.3 (Solution of a homogeneous first-order recurrence relation)

Let us suppose that the sequence $(a_n)_{n \in \mathbb{N}}$ satisfies the recurrence relation

$$a_n = Aa_{n-1}, \quad n \geq 2 \quad (8.2)$$

where $A \in \mathbb{R}$, and we know the initial condition a_1 . Then, the solution of this relation is given by

$$a_n = a_1 A^{n-1}, \quad n \geq 1 \quad (8.3)$$

Remark. In this course, we will only consider linear recurrence relations with constant coefficients (A in the previous theorem).

Theorem 8.4 (Solution of a homogeneous Fibonacci-type recurrence relation)

Let us suppose that the sequence $(a_n)_{n \in \mathbb{N}}$ satisfies the recurrence relation

$$a_n = Aa_{n-1} + Ba_{n-2}, \quad n \geq 3 \quad (8.4)$$

with $A, B \in \mathbb{R}$, and known initial conditions a_1, a_2 . If the **characteristic equation** associated to this relation

$$x^2 = Ax + B \quad (8.5)$$

has characteristic roots α and β , then the solution of the recurrence relation is given for all $n \geq 1$ by

$$a_n = \begin{cases} K_1 \alpha^n + K_2 \beta^n & \text{if } \alpha \neq \beta \\ (K_1 + nK_2) \alpha^n & \text{if } \alpha = \beta \end{cases} \quad (8.6)$$

where the constants K_1 and K_2 can be obtained using the initial conditions a_1, a_2 .

8.2.1 General Case

Let us suppose that the sequence $(a_n)_{n \in \mathbb{N}}$ satisfies the linear recursion:

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k}, \quad n \geq k+1 \quad (8.7)$$

with real numbers $c_1, c_2, \dots, c_k$. We assume that the k initial conditions $a_1, a_2, \dots, a_k$ are known.

If we look for a solution of the form

$$a_n = K_i x^n \quad (8.8)$$

then the amplitude K_i cancels out, and the indeterminate x should satisfy the **characteristic equation**:

$$x^k = c_1 x^{k-1} + c_2 x^{k-2} + \dots + c_k \quad (8.9)$$

If a_n and b_n are two solutions of a given linear homogeneous recurrence relation, then any linear combination $\alpha a_n + \beta b_n$ will be also a solution of that recursion.

The space of solutions has a vector space structure.

To each distinct characteristic root x_i , there corresponds a solution $a_n^{(i)}$, whose structure depends on the multiplicity of x_i :

- If the root x_i is simple, then $a_n^{(i)} = K_i x_i^n$.
- If the root x_i is double, then $a_n^{(i)} = (K_i + K_{i'} n) x_i^n$.
- If the root x_i is triple, then $a_n^{(i)} = (K_i + K_{i''} n + K_{i'''} n^2) x_i^n$, etc.

If the characteristic equation has r distinct roots x_i with multiplicities k_i (such that $\sum_{i=1}^r k_i = k$), then the general solution for this recursion has the form:

$$a_n = \sum_{i=1}^r \left[\sum_{j=1}^{k_i} K_i^{(j)} n^{j-1} \right] x_i^n, \quad n \geq 1 \quad (8.10)$$

where the k constants $K_i^{(j)}$ are determined using the k initial conditions.

8.3 Solution of a Linear Nonhomogeneous Recurrence Relation

Theorem 8.5 (Solution of a linear nonhomogeneous recurrence relation)

Let us assume that the sequence $(a_n)_{n \in \mathbb{N}}$ satisfies the linear nonhomogeneous recurrence relation with constant coefficients:

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k} + t_n, \quad n \geq k+1 \quad (8.11)$$

where $c_1, c_2, \dots, c_k \in \mathbb{R}$, and the initial conditions $a_1, \dots, a_k$ are known. The function $t_n : \mathbb{N} \rightarrow \mathbb{R}$ is a given **known function** of n . Then, the general solution of this linear nonhomogeneous recurrence is equal to the sum of the general solution for the linear homogeneous recurrence relation

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k}, \quad n \geq k+1 \quad (8.12)$$

plus any particular solution of the full recurrence.

Theorem 8.6 (Solution of a linear nonhomogeneous recurrence relation)

Let us suppose that the sequence $(a_n)_{n \in \mathbb{N}}$ satisfies the linear nonhomogeneous recurrence

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k} + t_n, \quad n \geq k+1 \quad (8.13)$$

where $c_1, c_2, \dots, c_k \in \mathbb{R}$, and the initial conditions $a_1, a_2, \dots, a_k$ are known. Let us further assume that the function $t_n : \mathbb{N} \rightarrow \mathbb{R}$ is of the form

$$t_n = s^n [b_0 + b_1 n + \dots + b_t n^t] \quad (8.14)$$

with real numbers $b_0, b_1, \dots, b_t$. If s is not a characteristic root of the associated linear homogeneous recurrence, then there exists a particular solution of the form

$$a_n^p = s^n [p_0 + p_1 n + \dots + p_t n^t] \quad (8.15)$$

If s is a characteristic root with multiplicity m of the associated linear homogeneous recurrence, then there exists a particular solution of the form

$$a_n^p = n^m s^n [p_0 + p_1 n + \dots + p_t n^t] \quad (8.16)$$

The particular solution a_n^p has no free parameters: there is only a unique choice for the coefficients $\{p_k\}_{k=1}^t$ such that a_n^p is actually a solution.

8.4 Generating Functions

- (i) **Recurrence relations.**
- (ii) **Generating functions:**

- Definitions.
- How to efficiently encode combinatorial problems?
- Solution of recurrence relations.

Definition 8.7

The **generating function** (GF) associated to the sequence $(a_k)_{k=0}^\infty$ is the following formal power series:

$$F(x) = a_0 + a_1 x + a_2 x^2 + \dots + a_n x^n + \dots = \sum_{n=0}^{\infty} a_n x^n \quad (8.17)$$

Examples:

- $(1+x)^k = \sum_{n=0}^k \binom{k}{n} x^n$ is the GF of $((\binom{k}{0}), (\binom{k}{1}), \dots, (\binom{k}{k}), 0, 0, \dots)$.
- $1 + x + x^2 + \dots + x^{k-1} = \sum_{n=0}^{k-1} x^n = \frac{1-x^k}{1-x}$ is the GF of $(1, 1, \dots, 1, 0, 0, \dots)$ where there are k ones.
- $\frac{1}{1-x} = 1 + x + x^2 + x^3 + \dots = \sum_{n=0}^{\infty} x^n$ is the GF of $(1, 1, 1, \dots)$.
- $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots = \sum_{n=0}^{\infty} \frac{x^n}{n!}$ is the GF of $(1, 1, \frac{1}{2!}, \frac{1}{3!}, \dots)$.

8.5 Basic Operations with Generating Functions

The GF for the sequence $(1, 2, 3, \dots)$ is given by

$$\sum_{n=0}^{\infty} (n+1) x^n = \frac{d}{dx} \sum_{n=0}^{\infty} x^{n+1} = \frac{d}{dx} \frac{x}{1-x} = \frac{1}{(1-x)^2} \quad (8.18)$$

If $F(x) = \sum_{n=0}^{\infty} a_n x^n$, and $G(x) = \sum_{n=0}^{\infty} b_n x^n$, then

$$(F + G)(x) = \sum_{n=0}^{\infty} (a_n + b_n) x^n \quad (8.19)$$

If F is the GF of the sequence $\{a_n\}_{n=0}^{\infty}$, then the GF of the sequence $(0, 0, \dots, 0, a_0, a_1, \dots)$ where there are k zeros is $G(x) = x^k F(x)$.

8.6 Integer Partitions

Problem 7: Count the number of distinct partitions of the positive integer N . For example, if $N = 4$, there are 5 partitions: $4 = 3 + 1 = 2 + 2 = 2 + 1 + 1 = 1 + 1 + 1 + 1$.

- (i) The sum principle allows us to compute the generating function associated to use the positive integer k in the partition:
 - The generating function for using 1 in the partition is $f_1 = 1 + x + x^2 + x^3 + \dots = \frac{1}{1-x}$.
 - The generating function for using $2 = 1 + 1$ in the partition is $f_2 = 1 + x^2 + x^4 + x^6 + \dots = \frac{1}{1-x^2}$.
 - The generating function for using $p \geq 1$ in the partition is $f_p = 1 + x^p + x^{2p} + x^{3p} + \dots = \frac{1}{1-x^p}$.
- (ii) Because writing up a partition is a sequential process, the generating function that encodes Problem 7 is given by the product principle:

$$f(x) = \prod_{k=1}^{\infty} f_k(x) = \prod_{k=1}^{\infty} \frac{1}{1-x^k} = 1 + x + 2x^2 + 3x^3 + 5x^4 + 7x^5 + \dots \quad (8.20)$$

8.7 Practical Procedure

Encoding of a combinatorial problem:

- (i) Compute the generating function F by using the sum/product principles and other operations.
- (ii) Compute the coefficients a_n by performing the Taylor power-series expansion of F around $x = 0$.

Solving a recurrence relation:

- (i) Rewrite the recurrence relation for a_n in terms of an equation that only involves the generating function F .
- (ii) Solve this equation and obtain a closed form for F in terms of x .
- (iii) Compute the coefficients a_n by performing the Taylor power-series expansion of F around $x = 0$.

8.8 Example: The Fibonacci Recursion

We want to solve the recurrence relation

$$f_n = f_{n-1} + f_{n-2}, \quad n \geq 2, \quad f_0 = 0, \quad f_1 = 1 \quad (8.21)$$

by using the generating function

$$F(x) = \sum_{n=0}^{\infty} f_n x^n = f_0 + f_1 x + \sum_{n=2}^{\infty} f_n x^n \quad (8.22)$$

Algorithm:

- (i) Multiply the recurrence relation by x^n , and sum over all values of n for which this recursion is valid (in our case, $n \geq 2$):

$$\sum_{n=2}^{\infty} f_n x^n = \sum_{n=2}^{\infty} f_{n-1} x^n + \sum_{n=2}^{\infty} f_{n-2} x^n \quad (8.23)$$

- (ii) Manipulate the sums so that they can be expressed in terms of F and the initial conditions $f_0 = 0, f_1 = 1$:

- $\sum_{n=2}^{\infty} f_n x^n = F - f_0 - f_1 x = F - x.$
- $\sum_{n=2}^{\infty} f_{n-1} x^n = x \sum_{n=2}^{\infty} f_{n-1} x^{n-1} = x \sum_{m=1}^{\infty} f_m x^m = x(F - f_0) = xF.$
- $\sum_{n=2}^{\infty} f_{n-2} x^n = x^2 \sum_{n=2}^{\infty} f_{n-2} x^{n-2} = x^2 \sum_{m=0}^{\infty} f_m x^m = x^2 F.$

The Fibonacci recursion now becomes the equation

$$F - x = xF + x^2 F \quad (8.24)$$

- (iii) We solve this equation for F :

$$F(x) = \frac{x}{1 - x - x^2} = \sum_{n=0}^{\infty} f_n x^n \quad (8.25)$$

- (iv) We compute the Taylor power-series expansion of F and we read the coefficient of x^n :

$$F(x) = \frac{x}{1 - x - x^2} = x + x^2 + 2x^3 + 3x^4 + 5x^5 + 8x^6 + 13x^7 + \dots \quad (8.26)$$

We can obtain all coefficients with a little algebra:

$$\begin{aligned} F(x) &= \frac{\alpha}{x + (1 + \sqrt{5})/2} + \frac{\beta}{x + (1 - \sqrt{5})/2} = \frac{1}{\sqrt{5}} \left[\frac{1}{1 - x(1 + \sqrt{5})/2} - \frac{1}{1 - x(1 - \sqrt{5})/2} \right] \\ &= \sum_{n=0}^{\infty} \frac{x^n}{\sqrt{5}} \left[\left(\frac{1 + \sqrt{5}}{2} \right)^n - \left(\frac{1 - \sqrt{5}}{2} \right)^n \right] \end{aligned} \quad (8.27)$$

Therefore,

$$f_n = \frac{1}{\sqrt{5}} \left[\left(\frac{1 + \sqrt{5}}{2} \right)^n - \left(\frac{1 - \sqrt{5}}{2} \right)^n \right] \quad (8.28)$$

8.9 Generalized Binomial Theorem

Theorem 8.8

Let k be a fixed positive integer, then we have formally that

$$\frac{1}{(1+x)^k} = \sum_{n=0}^{\infty} \binom{-k}{n} x^n \quad (8.29)$$

where for all $n \geq 0$ the above binomial coefficient is defined as

$$\binom{-k}{n} = \frac{-k(-k-1)(-k-2)\dots(-k-n+1)}{n!} = (-1)^n \binom{n+k-1}{n} \quad (8.31)$$

9 Graph Theory V

9.1 Combinatorial Problems on Graphs

- (i) **Undirected graphs.**
- (ii) **Algorithms in graph theory.**
- (iii) **Combinatorial problems on graphs:**
 - **Perfect matchings.**
 - **Proper colorings.**

9.2 Perfect Matchings

Definition 9.1

A **perfect matching** of a simple graph with $2n$ vertices is a spanning subgraph composed by n disjoint edges.

Remarks.

- All the vertices of G belong to the matching.
- Each vertex of G is incident with a single edge belonging to the matching.
- If G is **bipartite**, then we can prove more theorems.

Theorem 9.2

If G is a bipartite and regular graph with degree $d \geq 1$, then G contains a perfect matching.

9.3 Proper Colorings: Chromatic Polynomial

Definition 9.3

Let $G = (V, E)$ be a simple graph and let $q \geq 2$ be a natural number. The **chromatic polynomial** P_G is a polynomial such that $P_G(q)$ gives the number of distinct proper colorings of G with $q \in \mathbb{N}$ colors.

Theorem 9.4

If $G = (V, E)$ is a simple graph, $P_G(q)$ is a polynomial in q .

The proof is based on these two facts:

- If $G = (\{v\}, \emptyset)$, $P_G(q) = q$.
- The contraction-deletion theorem.

Theorem 9.5 (The contraction-deletion theorem)

If $G = (V, E)$ is a simple graph, and $e = \{x, y\} \in E$ with $x, y \in V$, then

$$P_G(q) = P_{G-e}(q) - P_{G/e}(q) \quad (9.1)$$

where $G - e$ is the graph obtained from G by deleting the edge e , and G/e is the graph obtained from G by contracting the edge e (i.e., by identifying vertices x and y , and eliminating possible multiple edges).

Theorem 9.6

If G is a disconnected graph with $k \geq 1$ connected components G_j , then

$$P_G(q) = \prod_{j=1}^k P_{G_j}(q) \quad (9.2)$$

Theorem 9.7

If G is a graph that can be split into two parts G_1 and G_2 such that $G_1 \cap G_2 = K_n$ for some $n \geq 1$, then

$$P_G(q) = \frac{P_{G_1}(q) \times P_{G_2}(q)}{P_{K_n}(q)} \quad (9.3)$$

- (i) If $G = K_n$, then $P_{K_n}(q) = q(q-1)\dots(q-n+1)$.
- (ii) If G is a tree with n vertices T_n , then $P_{T_n}(q) = q(q-1)^{n-1}$.

10 Binary Relations. Equivalence Relations

10.1 Binary Relations

- (i) **Binary relations:**
 - Definitions.
 - Graphical representation of a relation.
 - Operations with relations.
 - Properties.
- (ii) **Equivalence relations:**
 - Equivalence classes.
 - Quotient set.
- (iii) **Order relations.**
- (iv) **Lattices and Boolean algebras.**

10.2 Binary Relations Between Two Sets

Definition 10.1

A **binary relation** R between the sets V and W is a subset of the Cartesian product $V \times W$:

$$V \times W = \{(v, w) : (v \in V) \wedge (w \in W)\} \quad (10.1)$$

Therefore, $R \subseteq V \times W$. The **domain** of R is the set:

$$\text{Dom } R = \{v \in V : (v, w) \in R \text{ for some } w \in W\} \quad (10.2)$$

and the **image** of R is the set:

$$\text{Im } R = \{w \in W : (v, w) \in R \text{ for some } v \in V\} \quad (10.3)$$

Notation: If $(v, w) \in R$, we denote it as vRw .

10.3 Binary Relations on a Set

Definition 10.2

A **binary relation** R on the set V is a subset of the Cartesian product $V \times V$. Hence, $R \subseteq V \times V$. The **domain** of R is the set:

$$\text{Dom } R = \{v \in V : (v, w) \in R \text{ for some } w \in V\} \quad (10.4)$$

and the **image** of R is the set:

$$\text{Im } R = \{w \in V : (v, w) \in R \text{ for some } v \in V\} \quad (10.5)$$

Important remark: A function $f : A \rightarrow B$ is a relation between the sets A and B and such that to each element $x \in \text{Dom}(f)$ there corresponds a unique element of B (i.e., $f(x)$).

10.4 Graphical Representation of a Relation

- Cartesian representation.
- Venn's diagrams.
- **Adjacency matrix of R :** Let V and W be the sets $V = \{v_1, v_2, \dots, v_{|V|}\}$ and $W = \{w_1, w_2, \dots, w_{|W|}\}$. Then entry (i, j) of A_R is equal to 1 if $v_i R w_j$, and it is equal to 0 otherwise.
- **Directed graph G_R associated to R :** The vertices of G_R are the elements of the set V where the relation R is defined. The set of (directed) edges is the set of ordered pairs:

$$E = \{(v_i, v_j) \in V \times V : v_i R v_j\} \quad (10.6)$$

10.5 Operations with Relations

Definition 10.3

Given the relation R on V , we define its **inverse relation** R^{-1} as the relation on V defined as $(v_1, v_2) \in R^{-1} \iff (v_2, v_1) \in R$, or in other words, $v_1 R^{-1} v_2 \iff v_2 R v_1$.

Given any relation R , there always exists its inverse relation R^{-1} . On the other hand, the inverse function f^{-1} exists if and only if f is bijective.

Definition 10.4

Given a relation R on V , we define its **complementary relation** $\bar{R}$ as the relation on V such that $(v_1, v_2) \in \bar{R} \iff (v_1, v_2) \notin R$.

Binary relations are subsets of $V \times W$, therefore, set operations can be interpreted as operations with relations.

10.6 Composition of Relations

Definition 10.5

Let R be a relation between the sets V and W , and let S be a relation between the sets W and Y . The composition of the relation S and R is a relation between the sets V and Y denoted as $S \circ R$. In particular, $S \circ R$ is a subset of the Cartesian product $V \times Y$ such that, given any $v \in V$ and $y \in Y$, $v(S \circ R)y$ if and only if there exists some $w \in W$ satisfying vRw and wSy .

Proposition 10.6

Let A_R be the adjacency matrix of the relation R between V and W , and let A_S be the adjacency matrix of the relation S between W and Y . Then, the adjacency matrix $A_{S \circ R}$ of the composition of the relations $S \circ R$ between V and Y is given by:

$$A_{S \circ R} = A_R \circ A_S \quad (10.7)$$

where the product $\circ$ is the Boolean product of matrices.

Using Boolean operations (instead of regular ones) guarantees that $A_{S \circ R}$ is an adjacency matrix associated to a binary relation.

10.7 Properties of Relations on a Set V

Definition 10.7

A relation R is reflexive if for every $v \in V$, vRv .

Definition 10.8

A relation R is irreflexive if for every $v \in V$, vRv does not hold.

Definition 10.9

A relation R is symmetric if $R = R^{-1}$, i.e., if $vRw \implies wRv$.

Definition 10.10

A relation R is antisymmetric if $(v_1Rv_2) \wedge (v_2Rv_1) \implies v_1 = v_2$.

10.8 Transitive Relations

Definition 10.11

A relation R is transitive if $(v_1Rv_2) \wedge (v_2Rv_3) \implies v_1Rv_3$.

Proposition 10.12

A relation R is transitive if and only if $R^n \subseteq R$ for all $n \in \mathbb{N}$. The n -th power R^n of the relation R is recursively defined as follows:

$$R^1 = R, \quad R^n = R \circ R^{n-1} \quad (10.8)$$

Corollary 10.12.1

A relation R is transitive if and only if $R^2 \subseteq R$. In other words, R is transitive if and only if for each nonzero entry $(A_{R^2})_{i,j} = 1$ of the adjacency matrix of R^2 , the corresponding entry of the adjacency matrix of R is also nonzero $(A_R)_{i,j} = 1$.

10.9 Equivalence Relations

Definition 10.13

A relation R on a set V is an equivalence relation if it is reflexive, symmetric and transitive.

Notation: If R is an equivalence relation, aRb is usually denoted as $a \equiv b \pmod{R}$.

Definition 10.14

Let R be an equivalence relation on a set V . The set of all the elements of V related to a certain element $v \in V$ is called the equivalence class determined by v , and it is denoted as $[v]_R$, or simply as $[v]$. Therefore,

$$[v]_R = \{w \in V : vRw\} \quad (10.9)$$

Any element $w \in [v]_R$ (in particular, v) is a representative of the equivalence class $[v]_R$.

10.10 Quotient Set

Theorem 10.15

Let R be an equivalence relation on V . Then,

- (i) $[a]_R$ is non-empty for all $a \in V$.
- (ii) For any two elements $a, b \in V$, either $[a]_R = [b]_R$ (and aRb), or $[a]_R \cap [b]_R = \emptyset$.
- (iii) The equivalence classes determine the relation uniquely.

Theorem 10.16

Let R be an equivalence relation on V . Then the set of all equivalence classes of R form a partition of V . Conversely, given a partition $\{V_1, V_2, \dots\}$ of V , there exists an equivalence relation R such that its equivalence classes are the sets V_i .

Definition 10.17

Let R be an equivalence relation on V . The set of all the equivalence classes of R is called the quotient set of V by R , and it is denoted by V/R :

$$V/R = \{[v]_R : v \in V\} \quad (10.10)$$

11 Integer and Modular Arithmetic

This chapter covers:

- (i) Integer arithmetic:
 - Integer divisibility
 - Euclid's algorithm
 - Bézout's identity
 - Linear Diophantine equations
- (ii) Modular arithmetic:
 - Linear congruences
 - Arithmetic on $\mathbb{Z}_p$
 - Euler's φ function. Euler's theorem

11.1 Integer Divisibility

The set of integers $\mathbb{Z}$ is closed with respect to the operations of sum, subtraction, and product. In other words, for every $a, b \in \mathbb{Z}$, $a \pm b \in \mathbb{Z}$ and $a \cdot b \in \mathbb{Z}$. They also satisfy:

- 0 is the identity with respect to the sum: $a + 0 = a$ for every $a \in \mathbb{Z}$.
- 1 is the identity with respect to the product: $a \cdot 1 = a$ for every $a \in \mathbb{Z}$.
- For every $a \in \mathbb{Z}$, there exists a unique inverse element $-a \in \mathbb{Z}$ such that $a + (-a) = 0$.

However, the result of dividing two integers might not be an integer.

Definition 11.1

Given two integers $a \neq 0$ and b , we say that a divides b if there is an integer $q \in \mathbb{Z}$ such that $b = a \cdot q$. If a divides b , we say that a is a factor of b and that b is a multiple of a . We denote $a \mid b$ when a divides b , and we write $a \nmid b$ when a does not divide b .

Remarks:

- Every non-zero integer $a \in \mathbb{Z} \setminus \{0\}$ divides 0: $0 = a \cdot 0$ ($q = 0$).
- 1 divides any $a \in \mathbb{Z}$: $a = 1 \cdot a$ ($q = a$).
- Any nonzero integer $a \in \mathbb{Z} \setminus \{0\}$ divides itself: $a = a \cdot 1$ ($q = 1$).

11.2 The Division Algorithm

Theorem 11.2 (The Division Algorithm)

Let a and $b \neq 0$ be two integers. Then there exists a unique pair of integers q and r such that

$$a = q \cdot b + r \quad \text{with } 0 \leq r < |b| \quad (11.1)$$

- The numbers a and b are called dividend and divisor, respectively.
- The number r is the remainder: $r = a \bmod b$.
- The number q is the quotient: $q = a \operatorname{div} b = \begin{cases} \lfloor a/b \rfloor & \text{if } b > 0 \\ \lceil a/b \rceil & \text{if } b < 0 \end{cases}$

11.3 Properties of Integer Division

Theorem 11.3

Let a, b, c be integers. Then:

- (i) If $a \mid b$ and $a \mid c$, then $a \mid (b + c)$.
- (ii) If $a \mid b$, then $a \mid (b \cdot c)$ for every $c \in \mathbb{Z}$.
- (iii) If $a \mid b$ and $b \mid c$, then $a \mid c$.
- (iv) If $c \neq 0$, then $a \mid b$ if and only if $(c \cdot a) \mid (c \cdot b)$.
- (v) If $a \mid b$ and $b \neq 0$, then $|a| \leq |b|$.
- (vi) If $a \mid b$ and $b \mid a$, then $a = \pm b$.

Theorem 11.4

If $a \mid b_i$ for $i = 1, \dots, N$, then $a \mid \sum_{i=1}^N u_i \cdot b_i$ for every $u_i \in \mathbb{Z}$.

11.4 Greatest Common Divisor. Euclid's Lemma (IIIrd century BC)

Definition 11.5

Let a, b be integers, not both simultaneously zero. The largest integer d such that $d \mid a$ and $d \mid b$ is called the greatest common divisor of a and b . It is denoted by $\gcd(a, b)$.

Remarks:

- The case $a = b = 0$ is excluded because any integer divides 0.

- $\gcd(0, a) = |a|$ for every nonzero integer a .

Theorem 11.6

The greatest common divisor of two numbers is unique.

Definition 11.7

Two integers a and b are relatively prime if $\gcd(a, b) = 1$. The integers $a_1, a_2, \dots, a_n$ are pairwise relatively prime if $\gcd(a_i, a_j) = 1$ for any $1 \leq i < j \leq n$.

Lemma 11.8 (Euclid)

Given the integers a , $b \neq 0$, q and r , such that $a = q \cdot b + r$ with $0 \leq r < |b|$, then $\gcd(a, b) = \gcd(b, r)$.

11.5 Euclid's Algorithm

Problem 9: Apply recursively Euclid's lemma to compute $\gcd(662, 414)$.

$$a = b \cdot q + r \quad (11.2)$$

$$662 = 414 \cdot 1 + 248 \quad (11.3)$$

$$414 = 248 \cdot 1 + 166 \quad (11.4)$$

$$248 = 166 \cdot 1 + 82 \quad (11.5)$$

$$166 = 82 \cdot 2 + 2 \quad (11.6)$$

$$82 = 2 \cdot 41 + 0 \quad (11.7)$$

$$\gcd(662, 414) = \gcd(414, 248) = \gcd(248, 166) = \gcd(166, 82) = \gcd(82, 2) = 2.$$

In general, $\gcd(a, b) = \gcd(b, r_1) = \gcd(r_1, r_2) = \dots = \gcd(r_{n-2}, r_{n-1})$, where r_{n-1} is the last nonzero remainder ($r_n = 0$). In the last step:

$$r_{n-2} = q_n \cdot r_{n-1} \implies r_{n-1} \mid r_{n-2} \quad (11.8)$$

Therefore, $\gcd(r_{n-2}, r_{n-1}) = r_{n-1}$.

Theorem 11.9

In Euclid's algorithm, $\gcd(a, b) = r_{n-1}$ (= the last nonzero remainder).

11.6 Bézout's Identity

Theorem 11.10 (Bézout's Identity, 1730-1783)

If a and b are two integers not simultaneously zero, then there exist integers u, w such that

$$\gcd(a, b) = a \cdot u + b \cdot w \quad (11.9)$$

Proof: We write the steps of Euclid's algorithm, and "unroll them":

$$a = q_1 \cdot b + r_1 \implies r_1 = a - q_1 \cdot b \quad (11.10)$$

$$b = q_2 \cdot r_1 + r_2 \implies r_2 = b - q_2 \cdot r_1 \quad (11.11)$$

$$r_1 = q_3 \cdot r_2 + r_3 \implies r_3 = r_1 - q_3 \cdot r_2 \quad (11.12)$$

$$\vdots \quad (11.13)$$

$$r_{n-4} = q_{n-2} \cdot r_{n-3} + r_{n-2} \implies r_{n-2} = r_{n-4} - q_{n-2} \cdot r_{n-3} \quad (11.14)$$

$$r_{n-3} = q_{n-1} \cdot r_{n-2} + r_{n-1} \implies r_{n-1} = r_{n-3} - q_{n-1} \cdot r_{n-2} \quad (11.15)$$

$$r_{n-2} = q_n \cdot r_{n-1} + (r_n = 0) \implies r_{n-1} = \gcd(a, b) \quad (11.16)$$

Then,

$$\gcd(a, b) = r_{n-1} = \alpha_{n-1}r_{n-3} + \beta_{n-1}r_{n-2} \quad (11.17)$$

$$= \alpha_{n-2}r_{n-4} + \beta_{n-2}r_{n-3} \quad (11.18)$$

$$= \dots \quad (11.19)$$

$$= \alpha_3r_1 + \beta_3r_2 \quad (11.20)$$

$$= \alpha_2b + \beta_2r_1 \quad (11.21)$$

$$= \alpha_1a + \beta_1b \quad (11.22)$$

Important remark: Bézout's identity does not imply that the integers u, v are unique.

Theorem 11.11

Let a and b be two integers not simultaneously zero with $\gcd(a, b) = d$. An integer c can be written in the form $a \cdot x + b \cdot y$ for some integers x, y if and only if c is a multiple of d . In particular, d is the smallest positive integer of the form $a \cdot x + b \cdot y$ with $x, y \in \mathbb{Z}$.

Corollary 11.11.1

Two integers are relatively prime if and only if there exist integers x, y such that $a \cdot x + b \cdot y = 1$.

Corollary 11.11.2

If $\gcd(a, b) = d$, then

- (i) $\gcd(m \cdot a, m \cdot b) = m \cdot d$ for every $m \in \mathbb{N}$.
- (ii) $\gcd(a/d, b/d) = 1$.

Corollary 11.11.3

If a, b are two relatively-prime integers, then:

- (i) If $a \mid c$ and $b \mid c$, then $(a \cdot b) \mid c$.
- (ii) If $a \mid (b \cdot c)$, then $a \mid c$.

11.7 Least Common Multiple

Definition 11.12

The least common multiple of two natural numbers a, b is the least natural number m such that $a \mid m$ and $b \mid m$. It is denoted by $\text{lcm}(a, b)$.

Remark: This number exists because the set of natural numbers $\mathbb{N}$ is a well-ordered set (see next chapter).

Theorem 11.13

If a, b are two natural numbers, then

$$\gcd(a, b) \cdot \text{lcm}(a, b) = a \cdot b \quad (11.23)$$

11.8 Prime Numbers

Definition 11.14

A natural number $p > 1$ is called a prime number if the only positive factors of p are 1 and p . A natural number $p > 1$ that is not prime is called composite.

Remark: The natural number 1 is not prime. The first prime number is 2, and the other prime numbers are odd natural numbers (3, 5, 7, 11, ...).

Theorem 11.15 (Euclid)

There are infinitely many prime numbers.

Theorem 11.16

The positive integer n is a composite number if and only if n can be divided by some prime number $p \leq \sqrt{n}$.

Lemma 11.17

Let p be a prime number, and let a, b be integers. Then: (a) Either $p \mid a$, or p and a are relatively prime. (b) If $p \mid (a \cdot b)$, then either $p \mid a$ or $p \mid b$.

11.9 The Fundamental Theorem of Arithmetic

Theorem 11.18 (The Fundamental Theorem of Arithmetic)

Every natural number $n > 1$ can be written uniquely as a product of primes

$$n = p_1^{n_1} \cdot p_2^{n_2} \cdot p_3^{n_3} \cdot \dots \cdot p_k^{n_k} \quad (11.24)$$

where the p_i are distinct prime numbers written in increasing order, and the exponents n_i are natural numbers $n_i \geq 1$.

Proposition 11.19

If the integers $a, b > 1$ can be factorized in the form

$$a = p_1^{n_1} \cdot p_2^{n_2} \dots p_k^{n_k} \quad (11.25)$$

$$b = p_1^{m_1} \cdot p_2^{m_2} \dots p_k^{m_k} \quad (11.26)$$

with $n_i, m_i \geq 0$ and all prime factors of a and b appear in both decompositions, then

$$\gcd(a, b) = p_1^{\min(n_1, m_1)} \cdot p_2^{\min(n_2, m_2)} \dots p_k^{\min(n_k, m_k)} \quad (11.27)$$

$$\text{lcm}(a, b) = p_1^{\max(n_1, m_1)} \cdot p_2^{\max(n_2, m_2)} \dots p_k^{\max(n_k, m_k)} \quad (11.28)$$

11.10 Linear Diophantine Equations [Diophantus of Alexandria, IIIrd century]

Definition 11.20

A Diophantine equation is an equation of one or several variables such that we are only interested in their integer solutions.

Theorem 11.21 (Brahmagupta, VIIth century)

The linear equation

$$a \cdot x + b \cdot y = c \quad (11.29)$$

where a, b, c are integers (and a, b not simultaneously zero), admits integer solutions if and only if $d = \gcd(a, b)$ divides c . In this case, there exist infinitely many integer solutions (x_k, y_k) with $k \in \mathbb{Z}$ given by

$$x_k = u \cdot p + (b \cdot k)/d \quad (11.30)$$

$$y_k = w \cdot p - (a \cdot k)/d \quad (11.31)$$

where $p = c/d \in \mathbb{Z}$ and u, w are given by

$$d = u \cdot a + w \cdot b \quad (11.32)$$

11.11 Modular Arithmetic

Modular arithmetic allows us to perform algebraic operations using, instead of a given set of numbers, their respective remainders with respect to some fixed positive number called the modulus. The modulus is 12 or 24 when we count hours with a clock, 7 when we count days in a week, etc.

Definition 11.22

Let a, b be integers, and let m be a natural number. Then a, b are congruent modulo m if $m \mid (a - b)$. This relation is denoted as $a \equiv b(\text{mod } m)$.

Proposition 11.23

- (i) $a \equiv b(\text{mod } m)$ if and only if $a \text{ mod } m = b \text{ mod } m$.
- (ii) $a \equiv b(\text{mod } m)$ if and only if $a = b + k \cdot m$ for some $k \in \mathbb{Z}$.

Theorem 11.24

For each positive integer m , the binary relation $\equiv (\text{mod } m)$ is an equivalence relation.

11.12 The Quotient Set $\mathbb{Z}_m$

The equivalence classes (or congruence classes) modulo m

$$[a]_m = \{b \in \mathbb{Z} : a \equiv b(\text{mod } m)\} = \{a + mk : k \in \mathbb{Z}\} \quad (11.33)$$

form a partition of $\mathbb{Z}$. There are m distinct equivalence classes corresponding to the m possible remainders obtained by dividing an integer by m .

Theorem 11.25

The quotient set $\mathbb{Z}_m = \mathbb{Z} / \equiv (\text{mod } m)$ is given by

$$\mathbb{Z}_m = \{[a]_m : 0 \leq a \leq m - 1\} \quad (11.34)$$

Remark: Usually, the notation for $\mathbb{Z}_m$ is a bit sloppy:

$$\mathbb{Z}_m = \{0, 1, 2, \dots, m - 1\} \quad (11.35)$$

11.13 Modular Arithmetic

Theorem 11.26

Let m be a positive integer. If $a_1 \equiv b_1(\text{mod } m)$ and $a_2 \equiv b_2(\text{mod } m)$, then:

- $a_1 \pm a_2 \equiv b_1 \pm b_2(\text{mod } m)$.
- $a_1 \cdot a_2 \equiv b_1 \cdot b_2(\text{mod } m)$.

Corollary 11.26.1

Let m, k be positive integers, and let $a, b \in \mathbb{Z}$. If $a \equiv b(\text{mod } m)$, then $a^k \equiv b^k(\text{mod } m)$.

Theorem 11.27

Let m be a positive integer, and let a, b, c be integers. If $a \cdot c \equiv b \cdot c \pmod{m}$ and $\gcd(c, m) = 1$, then $a \equiv b \pmod{m}$.

Remarks:

- This theorem allows us to divide by a common factor c both sides of the sign $\equiv$ whenever c and the modulus m are relatively prime.
- If c and m are not relatively prime, then the correct result is: Let us write $m = p \cdot c$ for positive integers p, c , and let a, b be integers. If $a \cdot c \equiv b \cdot c \pmod{p \cdot c}$, then $a \equiv b \pmod{p}$.

11.14 Modular Division: Linear Congruence Equations

Definition 11.28

A congruence modulo m of the form

$$a \cdot x \equiv b \pmod{m} \quad (11.36)$$

where m is a positive integer, a, b are integers, and x is a variable is called a linear congruence equation.

Remarks:

- If there exists a unique solution of the linear congruence equation $a \cdot x \equiv 1 \pmod{m}$, then solving this equation is equivalent to obtaining the multiplicative inverse of a modulo m .
- If x is a solution of a linear congruence equation, and $x' \equiv x \pmod{m}$, then x' is also a solution of that equation: $a \cdot x' \equiv a \cdot x \pmod{m} \equiv b \pmod{m}$.
- Therefore, the solutions of a linear congruence equation (if any) form classes of congruence modulo m : i.e., they are elements of $\mathbb{Z}_m$.

11.15 Linear Congruence Equations

Theorem 11.29

If $d = \gcd(a, m)$, then the linear congruence equation

$$a \cdot x \equiv b \pmod{m} \quad (11.37)$$

has a solution if and only if $d \mid b$. In this case and if x_0 is a particular solution of the linear congruence equation, the general solution is given by

$$x_k = x_0 + (m \cdot k)/d, \quad k \in \mathbb{Z} \quad (11.38)$$

In particular, these solutions form d congruence classes modulo m with representatives: $\{x_0, x_0 + m/d, x_0 + (2m)/d, \dots, x_0 + (m(d-1))/d\}$

Corollary 11.29.1

If $\gcd(a, m) = 1$, the solutions x of the linear congruence equation $a \cdot x \equiv b \pmod{m}$ form a unique congruence class modulo m .

Corollary 11.29.2

If $\gcd(a, m) = 1$ with $m > 1$, then there exists a multiplicative inverse of a modulo m . This multiplicative inverse is unique modulo m .

11.16 Arithmetic with $\mathbb{Z}_m$

The elements of $\mathbb{Z}_m$ with $m \in \mathbb{N}$ are equivalence classes modulo m . For the sake of simplicity, $x \in \mathbb{Z}_m$ represents that $x \in [x]_m$.

The sum and the multiplication on $\mathbb{Z}_m$ are defined as:

$$x + y = [x]_m + [y]_m = [x + y]_m \quad (11.39)$$

$$x \cdot y = [x]_m \cdot [y]_m = [x \cdot y]_m \quad (11.40)$$

and they verify the usual properties: for every $x, y, z \in \mathbb{Z}_m$,

- Closure: $x + y \in \mathbb{Z}_m$ and $x \cdot y \in \mathbb{Z}_m$.
- Associativity: $x + (y + z) = (x + y) + z$ and $x \cdot (y \cdot z) = (x \cdot y) \cdot z$.
- Commutativity: $x + y = y + x$ and $x \cdot y = y \cdot x$.
- Distributivity: $x \cdot (y + z) = x \cdot y + x \cdot z$.
- Identity element (sum): $\exists 0 \in \mathbb{Z}_m$ such that $0 + x = x$, $\forall x \in \mathbb{Z}_m$.
- Identity element (product): $\exists 1 \in \mathbb{Z}_m$ such that $1 \cdot x = x$, $\forall x \in \mathbb{Z}_m$.
- Inverse element (sum): $\forall x \in \mathbb{Z}_m$, $\exists -x \in \mathbb{Z}_m$ such that $x + (-x) = 0$.

Remark: These properties are those characterizing a field [like $(\mathbb{R}, +, \cdot)$], except for the existence of a multiplicative inverse.

In $\mathbb{Z}$ there does not exist in general the multiplicative inverse of an integer x : y is the multiplicative inverse of x if and only if $x \cdot y = 1$. However, two properties hold:

- (i) Cancellation law: If $x \neq 0$ and $x \cdot y = x \cdot z$, then $y = z$.
- (ii) If $x \cdot y = 0$, then either $x = 0$ or $y = 0$.

None of these two properties holds in general in $\mathbb{Z}_m$.

Definition 11.30

An element $x \neg \equiv 0 \pmod{m}$ of $\mathbb{Z}_m$ is a divisor of zero if there exists an element $y \neg \equiv 0 \pmod{m}$ such that $x \cdot y \equiv 0 \pmod{m}$.

Remark: In some books, the condition $x \neg \equiv 0 \pmod{m}$ is dropped.

Definition 11.31

An element $x \in \mathbb{Z}_m$ is a unit modulo m if it has a multiplicative inverse modulo m ; i.e., if there is an element $s \in \mathbb{Z}_m$ such that $x \cdot s \equiv 1 \pmod{m}$.

Theorem 11.32

The multiplicative inverse of a unit modulo m is unique.

Remark: As the inverse of a unit r modulo m is unique, it will be denoted by r^{-1} .

Theorem 11.33

An element $r \in \mathbb{Z}_m$ is invertible (i.e., it has a multiplicative inverse) if and only if r and m are relatively prime.

Corollary 11.33.1

If p is a prime number, every nonzero element of $\mathbb{Z}_p$ is invertible.

- If p is prime, then $(\mathbb{Z}_p, +, \cdot)$ is a field like $(\mathbb{R}, +, \cdot)$ or $(\mathbb{Q}, +, \cdot)$.
- If $m = p \cdot q$ is composite, then there are divisors of zero in $\mathbb{Z}_m$: $p \cdot q \equiv 0 \pmod{m}$ with $p, q \not\equiv 0 \pmod{m}$. In this case, $(\mathbb{Z}_m, +, \cdot)$ is a ring with divisors of zero.

Definition 11.34

Euler's (totient) function $\varphi : \mathbb{N} \rightarrow \mathbb{N}$ is defined as $\varphi(m)$ gives the number of invertible elements of $\mathbb{Z}_m$.

Lemma 11.35

If p is a prime number, then $\varphi(p) = p - 1$.

11.17 Euler's Theorem

Theorem 11.36 (Euler, 1790)

If y is invertible in $\mathbb{Z}_m$ (i.e., if $\gcd(y, m) = 1$), then

$$y^{\varphi(m)} \equiv 1 \pmod{m} \quad (11.41)$$

Corollary 11.36.1 (Fermat's Little Theorem)

If p is a prime number and if $y \not\equiv 0 \pmod{p}$, then

$$y^{p-1} \equiv 1 \pmod{p} \quad (11.42)$$

Corollary 11.36.2

If p is a prime number, then $y^p \equiv y \pmod{p}$ for any integer y .

Theorem 11.37

- (i) If p is a prime, then $\varphi(p^k) = p^{k-1}(p-1)$ for every $k \in \mathbb{N}$.
- (ii) If $\gcd(m, n) = 1$, then $\varphi(m \cdot n) = \varphi(m) \cdot \varphi(n)$.
- (iii) If $n \geq 2$ has the following decomposition in prime factors $n = \prod_{k=1}^r p_k^{n_k}$ with $n_k \geq 1$, then $\varphi(n) = n \cdot \prod_{k=1}^r (1 - 1/p_k)$.

12 Order Relations

This chapter covers:

- (i) Binary relations.
- (ii) Equivalence relations.
- (iii) Order relations:
 - Partially ordered sets.
 - Hasse diagrams.
 - Maximal elements.
 - Totally ordered sets.
 - Well-ordered sets and mathematical induction.
- (iv) Lattices and Boolean algebras.

12.1 Partial Order Relations

Definition 12.1

A binary relation on a set V is a partial order (or an order relation) if it is reflexive, antisymmetric, and transitive.

Notation: Order relations are usually denoted by the symbol $\preceq$.

Definition 12.2

A set V equipped with an order relation $\preceq$ is called a partially ordered set $(V, \preceq)$ (or poset).

Definition 12.3

Let $(V, \preceq)$ be a partially ordered set. Two elements $a, b \in V$ are comparable if either $a \preceq b$ or $b \preceq a$. If none of these conditions holds, such elements are incomparable.

Definition 12.4

A partially ordered set $(V, \preceq)$ is totally ordered when any pair of elements $a, b \in V$ are comparable. In this case, $(V, \preceq)$ is a totally ordered set (or linear order or chain).

12.2 Hasse Diagrams (1926)

The directed graph associated to an order relation $\preceq$ can be simplified by eliminating redundant elements.

Algorithm to obtain the Hasse diagram for a partial order $\preceq$:

- (i) As $\preceq$ is reflexive, there is a loop incident with each vertex. We eliminate all these loops.
- (ii) The transitivity of $\preceq$ implies the existence of subgraphs of the following type: If $a \preceq b$ and $b \preceq c$, we eliminate the superfluous edge associated to $a \preceq c$.
- (iii) We choose that all the oriented edges point upwards. Then, we eliminate all the arrows.

12.3 Extremal Elements

Definition 12.5

Let $(V, \preceq)$ be a partially ordered set. $M \in V$ is a maximal element if for all $v \in V$, $M \preceq v$ implies that $M = v$. $m \in V$ is a minimal element if for all $v \in V$, $v \preceq m$ implies that $m = v$. In other words, in the Hasse diagram associated to $(V, \preceq)$, there is no element above M , and no element below m .

Definition 12.6

Let $(V, \preceq)$ be a partially ordered set. $M^* \in V$ is a maximum (or greatest element) if $v \preceq M^*$ for all $v \in V$. $m^* \in V$ is a minimum (or least element) if $m^* \preceq v$ for all $v \in V$.

In other words, in the Hasse diagram associated to $(V, \preceq)$, M^* is above all the elements of V , and m^* is below all elements of V . The maximum and minimum of $(V, \preceq)$ are denoted by $\max(V)$ and $\min(V)$, respectively.

Remark: The maximal, minimal, greatest, and/or least elements of $(V, \preceq)$ might not exist.

Theorem 12.7

The maximum M^* of a partially ordered set $(A, \preceq)$, if it exists, is unique. In addition, the maximum of $(A, \preceq)$ is also a maximal element of it.

Definition 12.8

Let $(V, \preceq)$ be a partially ordered set, and $B \subset V$. $u \in V$ is an upper bound of B if $b \preceq u$ for all $b \in B$. The set of the upper bounds of B is denoted by $\text{major}(B)$.

The supremum of B , $\sup(B)$, is the least upper element of B : $\sup(B) = \min(\text{major}(B))$.

$d \in V$ is a lower bound of B if $d \preceq b$ for all $b \in B$. The set of all the lower bounds of B is denoted by $\text{minor}(B)$.

The infimum of B , $\inf(B)$, is the greatest lower element of B : $\inf(B) = \max(\text{minor}(B))$.

Remark: It may happen that $\text{major}(B) = \emptyset$, $\text{minor}(B) = \emptyset$ and/or $\sup(B)$ and $\inf(B)$ do not exist.

12.4 Total Order Compatible with a Partial Order

Definition 12.9

A total order $(V, \preceq_T)$ is compatible with the partial order $(V, \preceq_P)$ if for all $v, w \in V$, $v \preceq_P w$ implies that $v \preceq_T w$.

Algorithm 12.4: Topological Sort

```
1 k = 1
2 while V ≠ ∅
3   | v_k = a minimal element of (V, ⪯_P)
4   | V ← V \ {v_k}
5   | k ← k + 1
6 v_1 ⪯_T v_2 ⪯_T ... ⪯_T v_n is a total order compatible with (V, ⪯_P).
```

12.5 Well-Ordered Sets

Definition 12.10

$(V, \preceq)$ is a well-ordered set if $(V, \preceq)$ is a total order and any nonempty subset of V always has a minimum.

Remarks:

- The set of natural numbers with the usual order $(\mathbb{N}, \leq)$ is a well-ordered set. This property is equivalent to the induction principle.
- The totally-ordered set $(\mathbb{Z}, \leq)$ is not a well-ordered set; but as $\mathbb{Z}$ is isomorphic to $\mathbb{N}$, we can choose another order $\preceq$ such that $(\mathbb{Z}, \preceq)$ is a well-ordered set.

12.6 The Induction Principle for the Natural Numbers

Definition 12.11 (Induction Principle: Weak Version)

Let P be some property that satisfies the following conditions:

- (i) Base step: $P(1)$ is true.
- (ii) Inductive step: If $P(k)$ is true for an arbitrary and fixed k , then $P(k + 1)$ is true.

Then, $P(n)$ is true for every $n \in \mathbb{N}$.

Remark: The hypothesis in the inductive step ($P(k)$ is true) is called the induction hypothesis. To perform the inductive step, one assumes the induction hypothesis, and then uses this assumption to prove that $P(k + 1)$ is true.

Definition 12.12 (Induction Principle: Strong Version)

Let P be some property that satisfies the following conditions:

- (i) Base step: $P(1)$ is true.

- (ii) Inductive step: Given an arbitrary fixed k , if $P(m)$ is true for any $1 \leq m \leq k$, then $P(k + 1)$ is true.

Then, $P(n)$ is true for every $n \in \mathbb{N}$.

Proposition 12.13 (Strong Induction Principle for Well-Ordered Sets)

Let $(V, \preceq)$ be a well-ordered set, and P be some property that satisfies the following conditions:

- (i) Base step: $P(v_0)$ is true for $v_0 = \min(V)$.
(ii) Inductive step: Let w be an arbitrary fixed element of V , and let v be its successor. If $P(x)$ is true for all $v_0 \preceq x \preceq w$, then $P(v)$ is true.

Then, $P(v)$ is true for every $v \in V$.

12.7 Summary: Types of Relations

Relation	Reflexive	Symmetric	Antisymmetric	Transitive	Additional Properties
Equivalence	✓	✓	✗	✓	
Order	✓	✗	✓	✓	
Total order	✓	✗	✓	✓	Every pair is comparable
Well-ordered set	✓	✗	✓	✓	Every non-empty subset has a minimum

13 Lattices and Boolean Algebras

This chapter covers:

- (i) Binary relations.
- (ii) Equivalence relations.
- (iii) Order relations.
- (iv) Lattices and Boolean algebras:
 - Definitions and properties.
 - Bounded lattices.
 - Distributive lattices.
 - Complemented lattices.
 - Boolean algebras.

13.1 Lattices

Definition 13.1

A lattice is a nonempty partially ordered set $(A, \preceq)$ in which $\sup(\{a, b\})$ and $\inf(\{a, b\})$ exist for all $a, b \in A$.

- If $\sup(a, b)$ and $\inf(a, b)$ exist, they are unique.
- If $(A, \preceq)$ is a lattice, both operations can be considered as binary operations on A :
 - Their supremum is denoted by $\sup(a, b) = a \vee b \in A$.
 - Their infimum is denoted by $\inf(a, b) = a \wedge b \in A$.
- Not every partially ordered set is a lattice.
- A totally-ordered set is a lattice with $\sup(a, b) = \max(a, b)$ and $\inf(a, b) = \min(a, b)$.

13.2 Duality

- If $(A, \preceq)$ is a partially ordered set, then $(A, \succeq)$ is also a partially ordered set. The Hasse diagram of $(A, \succeq)$ is obtained by inverting the Hasse diagram of $(A, \preceq)$.
- If $(A, \preceq)$ is a lattice, then $(A, \succeq)$ is also a lattice, with the interchange $\sup \leftrightarrow \inf$.

Corollary 13.1.1 (Duality Principle)

Any statement about a lattice $(A, \preceq)$ is still valid if we make the interchanges $\preceq \leftrightarrow \succeq$, $\sup \leftrightarrow \inf$, and $\vee \leftrightarrow \wedge$.

- The lattices $(A, \preceq)$ and $(A, \succeq)$ are dual.
- The order relations $\preceq$ and $\succeq$ are dual.
- The operations $\vee$ and $\wedge$ are dual.

13.3 Lattice Properties

Proposition 13.2

If $(A, \preceq)$ is a lattice, then for any $a, b, c \in A$:

- (i) $\sup(a, a) = a \vee a = a$ [idempotent law].
- (ii) $\sup(a, b) = a \vee b = b \vee a = \sup(b, a)$ [commutativity law].
- (iii) $\sup(a, \sup(b, c)) = a \vee (b \vee c) = (a \vee b) \vee c = \sup(\sup(a, b), c)$ [associativity law].
- (iv) $\sup(a, \inf(a, b)) = a \vee (a \wedge b) = a$ [absorption law].

By duality, one obtains

Corollary 13.2.1

If $(A, \preceq)$ is a lattice, then for any $a, b, c \in A$:

- (i) $\inf(a, a) = a \wedge a = a$ [idempotent law].
- (ii) $\inf(a, b) = a \wedge b = b \wedge a = \inf(b, a)$ [commutativity law].
- (iii) $\inf(a, \inf(b, c)) = a \wedge (b \wedge c) = (a \wedge b) \wedge c = \inf(\inf(a, b), c)$ [associativity law].
- (iv) $\inf(a, \sup(a, b)) = a \wedge (a \vee b) = a$ [absorption law].

Proposition 13.3

If $(A, \preceq)$ is a lattice, then the following statements are equivalent for any $a, b \in A$:

- (i) $a \preceq b$.
- (ii) $\sup(a, b) = a \vee b = b$.
- (iii) $\inf(a, b) = a \wedge b = a$.

Proposition 13.4 (Distributive Inequalities)

If $(A, \preceq)$ is a lattice, then for any $a, b, c \in A$:

- (i) $\inf(a, \sup(b, c)) = a \wedge (b \vee c) \preceq (a \wedge b) \vee (a \wedge c) = \sup(\inf(a, b), \inf(a, c))$.
- (ii) $\sup(a, \inf(b, c)) = a \vee (b \wedge c) \succeq (a \vee b) \wedge (a \vee c) = \inf(\sup(a, b), \sup(a, c))$.

13.4 Lattices as Algebraic Structures

Definition 13.5

A lattice is an algebraic structure $(A, \vee, \wedge)$ with two binary operations $\vee$ and $\wedge$ that satisfy the commutative, associative, and absorption laws.

- The absorption law implies the idempotent law.
- Even though we do not assume the existence of any order relation on A , there is one order relation induced by the properties of the operations $\vee$ and $\wedge$. In particular, for any $a, b \in A$,

$$a \preceq b \iff a \vee b = b \quad (13.1)$$

- $a \preceq a$ because $a \vee a = a$ (idempotent law).
- If $a \preceq b \iff a \vee b = b$. If $b \preceq a \iff b \vee a = a$. Therefore, $a = b$.
- If $a \preceq b \iff a \vee b = b$ and $b \preceq c \iff b \vee c = c$, then $a \vee c = a \vee (b \vee c) = (a \vee b) \vee c = b \vee c = c$. Therefore $a \preceq c$.
- In summary, $\preceq$ is a partial order relation and $(A, \preceq)$ is a partially ordered set.

Definition 13.6

Given a lattice $(A, \vee, \wedge)$, a sublattice $(M, \vee, \wedge)$ of $(A, \vee, \wedge)$ is given by a nonempty subset $M \subseteq A$ such that $(M, \vee, \wedge)$ is also a lattice using the same operations as those used in $(A, \vee, \wedge)$. (In other words, $(M, \vee, \wedge)$ should be closed under the binary operations $\vee$ and $\wedge$.)

Any lattice is a sublattice of itself.

13.5 Bounded Lattices

Definition 13.7

A lattice $(A, \preceq)$ has a lower bound, denoted by 0, if $0 \preceq a$ for all $a \in A$. A lattice has an upper bound denoted by 1, if $a \preceq 1$ for all $a \in A$. A lattice is bounded if it contains a lower bound 0 and an upper bound 1.

The bounds 0 and 1 satisfy the following properties for all $a \in A$:

- $\sup(a, 1) = a \vee 1 = 1$.
- $\inf(a, 1) = a \wedge 1 = a$.
- $\sup(a, 0) = a \vee 0 = a$.
- $\inf(a, 0) = a \wedge 0 = 0$.
- The upper bound 1 is the identity element for $\wedge$: $a \wedge 1 = a$, and it satisfies $a \vee 1 = 1$.
- The lower bound 0 is the identity element for $\vee$: $a \vee 0 = a$, and it satisfies $a \wedge 0 = 0$.
- In a bounded lattice, we can extend the duality principle by considering the interchange $0 \leftrightarrow 1$.
- Any finite lattice A is bounded: $1 = \sup(A)$ and $0 = \inf(A)$.

13.6 Distributive Lattices

Definition 13.8

A lattice $(A, \preceq)$ is a distributive lattice if for all $a, b, c \in A$,

$$\inf(a, \sup(b, c)) = a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c) = \sup(\inf(a, b), \inf(a, c)) \quad (13.2)$$

$$\sup(a, \inf(b, c)) = a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c) = \inf(\sup(a, b), \sup(a, c)) \quad (13.3)$$

This property is stronger than the distributive laws:

$$\inf(a, \sup(b, c)) = a \wedge (b \vee c) \preceq (a \wedge b) \vee (a \wedge c) = \sup(\inf(a, b), \inf(a, c)) \quad (13.4)$$

$$\sup(a, \inf(b, c)) = a \vee (b \wedge c) \succeq (a \vee b) \wedge (a \vee c) = \inf(\sup(a, b), \sup(a, c)) \quad (13.5)$$

Theorem 13.9

A lattice is distributive if and only if it does not contain a sublattice that is isomorphic to any of the following two lattices: N_5 (the “pentagonal lattice”) and M_3 (the “diamond lattice”).

13.7 Complemented Lattices

Definition 13.10

Let $(A, \vee, \wedge, 0, 1)$ be a bounded lattice. An element $a \in A$ has a complement $b \in A$ if $\sup(a, b) = a \vee b = 1$ and $\inf(a, b) = a \wedge b = 0$.

- The bounds 0 and 1 are complements of each other.
- If a is a complement of b , then b is a complement of a .
- An element $a \in A$ may have no complements, or it may have several ones.
- The unique complement of 1 is 0, and vice versa.

Definition 13.11

A bounded lattice $(A, \vee, \wedge, 0, 1)$ is complemented if for each $a \in A$ there is at least one complement.

Proposition 13.12

Let $(A, \vee, \wedge)$ be a distributive lattice. If an element $a \in A$ has a complement, then this element is unique.

If $(A, \vee, \wedge)$ is a distributive and complemented lattice, then each element $a \in A$ has a unique complement. This element will be denoted by $\bar{a}$.

13.8 Boolean Algebras

Definition 13.13 (Definition 1)

A Boolean algebra is a bounded, distributive and complemented lattice $(A, \vee, \wedge, \bar{}, 0, 1)$.

Definition 13.14 (Definition 2)

Let B be a nonempty set with at least two distinct elements $0, 1$. We define on B the following operations:

- The (binary) Boolean sum $(a, b) \rightarrow a + b \in B$.
- The (binary) Boolean multiplication $(a, b) \rightarrow a \cdot b \in B$.
- The (unary) complementation $a \rightarrow \bar{a} \in B$.

Then B is a Boolean algebra if the following properties hold for all $a, b, c \in B$:

- (i) $a + 0 = a$ [identity w.r.t. the sum].
- (ii) $a \cdot 1 = a$ [identity w.r.t. the multiplication].
- (iii) $a + b = b + a, a \cdot b = b \cdot a$ [commutativity laws].
- (iv) $a + (b + c) = (a + b) + c, a \cdot (b \cdot c) = (a \cdot b) \cdot c$ [associativity laws].
- (v) $a + (b \cdot c) = (a + b) \cdot (a + c), a \cdot (b + c) = (a \cdot b) + (a \cdot c)$ [distributive laws].
- (vi) $a + \bar{a} = 1, a \cdot \bar{a} = 0$ [complement laws].

13.9 Simple Boolean Algebra

- We can drop the symbol $\cdot$ in the Boolean multiplication $a \cdot b = ab$ whenever there is no confusion.
- The elements $0, 1 \in A$ do not have to be equal to the numbers $0, 1 \in \mathbb{Z}$.
- The Boolean operations $+$ and $\cdot$ do not have to coincide with the sum and multiplication of real numbers.

Let $(B, +, \cdot, \bar{}, 0, 1)$ be an algebra with $B = \{0, 1\}$ and the operations $+$, $\cdot$, and $\bar{}$ defined on B as follows:

$$1 \cdot 0 = 0 \cdot 1 = 0 \cdot 0 = 0 \quad (13.6)$$

$$1 \cdot 1 = 1 \quad (13.7)$$

$$1 + 1 = 1 + 0 = 0 + 1 = 1 \quad (13.8)$$

$$0 + 0 = 0 \quad (13.9)$$

$$\bar{1} = 0 \quad (13.10)$$

$$\bar{0} = 1 \quad (13.11)$$

Then $(B, +, \cdot, \bar{}, 0, 1)$ is a Boolean algebra, and it is the simplest one that exists: the Boolean algebra of two elements.

13.10 General Non-Trivial Boolean Algebras

Let A be a nonempty set. We now consider the power set $\mathcal{P}(A)$ with the order relation for every pair $B, C \subseteq A$:

$$B \preceq C \iff B \subseteq C \quad (13.12)$$

- The set $(\mathcal{P}(A), \preceq)$ is a partially ordered set.
- The set $(\mathcal{P}(A), \preceq)$ is a lattice. Given $B, C \subseteq A$, then
 - $\sup(B, C) = B \cup C \subseteq A$ ($\vee \Rightarrow \cup$).
 - $\inf(B, C) = B \cap C \subseteq A$ ($\wedge \Rightarrow \cap$).
- The identities are
 - $1 = A$.
 - $0 = \emptyset$.
- The set $(\mathcal{P}(A), \cup, \cap, \emptyset, A)$ is a distributive lattice.
- Each $B \subseteq A$ has a unique complement $\bar{B} = A \setminus B \subseteq A$.
- The set $(\mathcal{P}(A), \cup, \cap, \setminus, \emptyset, A)$ is a Boolean algebra.
- Practical use in probability theory.

13.11 Properties of a Boolean Algebra

Proposition 13.15

Let $(B, +, \cdot, \bar{}, 0, 1)$ be a Boolean algebra. Then, for all $a, b \in B$:

- (i) Idempotent laws: $a + a = a$ and $a \cdot a = a$.
- (ii) Dominance laws: $a + 1 = 1$ and $a \cdot 0 = 0$.
- (iii) Absorption laws: $a \cdot (a + b) = a$ and $a + a \cdot b = a$.
- (iv) De Morgan laws: $\overline{(a + b)} = \bar{a} \cdot \bar{b}$ and $\overline{(a \cdot b)} = \bar{a} + \bar{b}$.
- (v) Involution law: $\bar{\bar{a}} = a$.
- (vi) $\bar{1} = 0$ and $\bar{0} = 1$.

Definition 13.16

Given a statement in a Boolean algebra, its dual statement is obtained by interchanging $+ \leftrightarrow \cdot$ and $0 \leftrightarrow 1$ in the original statement.

Proposition 13.17

If a theorem is a consequence of the definitions of Boolean algebra, then the dual of the theorem is also a theorem.

Definition 13.18

Let $(B, +, \cdot, \neg, 0, 1)$ be a Boolean algebra. Then a subset $C \subseteq B$ is a Boolean subalgebra if $0, 1 \in C$, and it is closed under the same operations $+$, $\cdot$, $\neg$.